FOUR RUNNERS THAT ARE BARELY LONELY

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ABSTRACT. For positive integers $v_1, \ldots, v_n$ let $\mathrm{ML}(v_1, \ldots, v_n) = \max_t \min_i \|v_i t\|$, where $\|x\|$ is the distance to the nearest integer. Kravitz asked which values below $1/n$ occur. For $n = 4$, Jain and Kravitz showed that the values below $1/4$ are $m/(4m+2)$, $m \geq 2$, up to finitely many undetermined exceptions. We settle the four-runner case completely. A primitive quadruple has ML $< 1/4$ if and only if it is, up to order, $\{1, 2, 3, 4k\}$, or $\{x, y, x+y, x+2y\}$ with $\gcd(x, y) = 1$ and $y \not\equiv x, 2x \pmod 4$, or $\{1, 3, 4, 14\}$. The values below $1/4$ are exactly $m/(4m+2)$ with $m \geq 2$, $m \neq 3$, so $3/14$ is the only exception, and Conjecture 4.1 of Fan and Sun holds. A density lemma bounds the length of a quadruple on neither critical subtorus by 512, and an exact search over 800,816,099 quadruples completes the proof. The same method shows that the two-dimensional subtori of $(\mathbb{R}/\mathbb{Z})^4$ with loneliness below $2/5$ take exactly the three-runner values in that range, which between $1/3$ and $2/5$ are the fractions p/q with $1 \leq 2q - 5p \leq 4$. Hence the accumulation points of the four-runner spectrum in $(0, 2/5]$ are the three-runner values there, as Giri and Kravitz conjectured, and $9/25$ is not the loneliness of a two-dimensional subtorus. For every n, assuming the conjecture for $n-1$ and $n-2$ speeds, we prove $\mathrm{ML}(v) > 1/n - (n(n-1))^{n-2}/(2|v|)$, which is sharp in $|v|$, and that every n-tuple with $\mathrm{ML} < 1/n$ and $|v| \geq n(n(n-1))^{2n-4}$ lies on a two-dimensional subtorus of loneliness $1/n$. This answers a question of Cordella. For five speeds, a two-dimensional subtorus of $(\mathbb{R}/\mathbb{Z})^5$ with loneliness below $2/9$ is the product of a coordinate circle with the circle of $\{1, 2, 3, 4\}$ or $\{1, 3, 4, 7\}$. We also give infinitely many subtori whose maximizers lie off every circle with a repeated speed.

1. INTRODUCTION

Let $\|x\| = \min_{k \in \mathbb{Z}} |x - k|$. For a finite set V of positive integers (the speeds) put

$$\mathrm{ML}(V) = \max_{t \in \mathbb{R}} \min_{v \in V} \|vt\|.$$

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The Lonely Runner Conjecture of Wills [22] and Cusick [9] predicts $\mathrm{ML}(V) \geq 1/(|V|+1)$. It is known for $|V| \leq 6$ [10, 7, 4, 5, 2], Rosenfeld proved it for seven speeds by a computer assisted argument [18], and further computer assisted proofs for up to fourteen speeds have been announced recently [20, 19, 1]. The survey [17] describes the history and the many equivalent formulations.

Kravitz [15] proposed to study the whole *spectrum* of values of ML for a fixed number $n = |V|$ of speeds. The example $V = \{1, \dots, n\}$ shows that the value $1/(n+1)$ is attained, and Kravitz asked whether the values between $1/(n+1)$ and $1/n$ form a rigid set. He conjectured that they are exactly the numbers $s/(ns+1)$, $s \in \mathbb{N}$, and proved this for $n \leq 3$. Fan and Sun [11] showed that for $n = 4$ the conjecture fails: $\mathrm{ML}(8, 4s+3, 4s+11, 4s+19) = (2s+7)/(8s+30)$ for every $s \geq 0$. They proposed an amended conjecture, in which $s/(ns+k)$ is allowed for $1 \leq k \leq n$, and they conjectured (their Conjecture 4.1) that for $n = 4$ only $k \in \{1, 2\}$ occurs. Jain and Kravitz [14] then developed a theory of *relative spectra* and proved [14, Theorem 1.3] that the set of values of ML below $1/4$ for four speeds has finite symmetric difference with

$$\left\{ \tfrac{1}{4} - \tfrac{1}{8s+12} : s \geq 1 \right\} = \left\{ \tfrac{m}{4m+2} : m \geq 2 \right\}. \tag{1}$$

(They state the result for $D = \frac{1}{2} - \mathrm{ML}$ on the interval $(\frac{1}{4}, \frac{1}{2}]$, where the set reads $\frac{1}{4} + 1/\operatorname{Prog}(8, 12)$ with $\operatorname{Prog}(8, 12) = \{8s + 12 : s \geq 0\}$. The term $s = 0$ is $D = \frac{1}{3}$, that is $\mathrm{ML} = \frac{1}{6}$, which no quadruple attains because $\mathrm{ML} \geq \frac{1}{5}$ for four speeds, and we omit it.) They write that determining this finite symmetric difference "is itself a finite calculation that we have not attempted to carry out", and that numerical experiments suggest that there are no exceptional elements. The value $\frac{3}{14}$, which corresponds to $s = 2$, turns out to be absent.

Cordella [8] carried the relative spectrum analysis to five and six speeds, again up to finitely many exceptions. The reduction behind all of these results, [8, Lemma 2.4], says that all but finitely many near-tight tuples lie on a two-dimensional "critical" subtorus, and it is not effective: no bound on the exceptional tuples is produced. Cordella asks [8, Question 6.7] for an effective version, or for a direct argument that bounds the speeds of a near-tight tuple lying on no critical subtorus. For four speeds he searched all quadruples with speeds at most 200 [8, Section 6.3] and found exactly one near-tight quadruple on neither critical subtorus, namely $(1, 3, 4, 14)$ with $\mathrm{ML} = 4/17$. The purpose of this paper is to give the effective bound for four speeds, to carry out the Jain–Kravitz calculation completely, and to classify the speed

sets. There is exactly one exceptional value and exactly one sporadic quadruple.

Call a quadruple $V = \{v_1, v_2, v_3, v_4\}$ of distinct positive integers *primitive* if $\gcd(v_1, v_2, v_3, v_4) = 1$. Since $\mathrm{ML}(cV) = \mathrm{ML}(V)$, nothing is lost by this normalization. For integers $x, y \geq 1$ write

$$V(x, y) = \{x,\ y,\ x + y,\ x + 2y\}.$$

Theorem 1.1 (Classification)**.** *Let V be a primitive quadruple of distinct positive integers. Then $\mathrm{ML}(V) < 1/4$ if and only if one of the following holds.*

(a) *$V = \{1, 2, 3, 4k\}$ for some $k \geq 1$, and then $\mathrm{ML}(V) = k/(4k + 1)$.*
(b) *$V = V(x, y)$ with $\gcd(x, y) = 1$, $x \neq y$, $y \not\equiv x \pmod 4$ and $y \not\equiv 2x \pmod 4$, and then $\mathrm{ML}(V) = m/(4m + 2)$ with $m = m(x, y)$ given by (2) below.*
(c) *$V = \{1, 3, 4, 14\}$, and then $\mathrm{ML}(V) = 4/17$.*

The function $m(x, y)$ is explicit. Let $r = (y - x) \bmod 4$ and $r' = (y - 2x) \bmod 4$, both in $\{1, 2, 3\}$ under the hypotheses of (b), and set

$$m_1 = \begin{cases} \frac{x+y-1}{2}, & r = 1, \\ \frac{x+3y-2}{4}, & r = 2, \\ \frac{x+3y-1}{2}, & r = 3, \end{cases} \qquad m_3 = \begin{cases} \frac{2x+y-1}{2}, & r' = 1, \\ \frac{2x+3y-2}{4}, & r' = 2, \\ \frac{2x+3y-1}{2}, & r' = 3, \end{cases} \tag{2}$$

and $m(x, y) = \max(m_1, m_3)$. These are integers (Lemma 8.5). The only overlap between the three families is $V(2, 1) = \{1, 2, 3, 4\}$.

Theorem 1.2 (The spectrum below $1/4$)**.** *The set of values $\mathrm{ML}(V)$ over all quadruples V of distinct positive integers with $\mathrm{ML}(V) < 1/4$ is exactly*

$$\left\{ \frac{m}{4m + 2} : m \geq 2,\ m \neq 3 \right\} = \left\{ \frac{1}{5}, \frac{2}{9}, \frac{5}{22}, \frac{3}{13}, \frac{7}{30}, \frac{4}{17}, \frac{9}{38}, \dots \right\}.$$

Equivalently, apart from the term $\frac{1}{6}$ excluded by the bound $\mathrm{ML} \geq \frac{1}{5}$, the finite symmetric difference in [14, Theorem 1.3] *consists of the single value $3/14$, which is omitted.*

Corollary 1.3. (i) *If $\mathrm{ML}(v_1, \dots, v_4) < 1/4$ then $\mathrm{ML} = s/(4s + k)$ with $k \in \{1, 2\}$, as Fan and Sun conjectured* [11, Conjecture 4.1]*. Their amended spectrum conjecture holds for $n = 4$.*

(ii) (Chen [7]) *The tight quadruples are $\{1, 2, 3, 4\}$ and $\{1, 3, 4, 7\}$, up to scaling. Equivalently $\mathrm{ML}(V) = 1/5$ for a primitive quadruple only in these two cases.*

(iii) *Every primitive quadruple other than the two tight ones satisfies $\mathrm{ML}(V) \geq 2/9$.*

Item (ii) is a theorem of Chen [7], see also Barajas and Serra [3] and [17, Section 4]. It appears to have been overlooked in the recent spectrum literature: Kravitz [15, Section 13] lists it as an open question, and Jain and Kravitz write in [14] that complete lists of tight instances were available only for $n \in \{1, 2, 3, 5\}$. Cordella [8, Section 2.5] gives the reference. Theorem 1.1 recovers Chen's result with a different proof. The problem of describing all tight instances for general n [17, Problem 1] remains open. See Goddyn and Wong [13] and Zhang [23] for structured families.

The same method goes one level up. Let $F(n_1, n_2) = \{(a, b, a + b, n_1 a + n_2 b)\}$ for $(n_1, n_2) \in \mathbb{Z}^2 \setminus \{0\}$, a two-dimensional subtorus of $\mathbb{T}^4$ (so $F(1, 2) = U^2$), and let $T_{(a,b,c)} \times \mathbb{T}$ denote the product of the circle of a triple with a free coordinate.

Theorem 1.4 (Two-dimensional subtori below 1/3)**.** *A coordinate-free two-dimensional subtorus U of $\mathbb{T}^4$ has $\mathrm{ML}(U) < 1/3$ if and only if, up to signed coordinate permutations, U is a product $T_{(a,b,c)} \times \mathbb{T}$ with $\mathrm{ML}(a, b, c) < 1/3$, or $U = F(n_1, n_2)$ with $n_1 + n_2 \equiv 0 \pmod 3$. In both cases $\mathrm{ML}(U) = s/(3s + 1)$ for some $s \geq 1$, with $3s = \max(|n_1 + n_2|, |2n_1 - n_2|, |2n_2 - n_1|)$ in the second case. Consequently the accumulation points of the four-runner spectrum in $(0, \frac{1}{3})$ are exactly the three-runner values $s/(3s + 1)$.*

The last sentence is the case $n = 4$ of a conjecture of Giri and Kravitz [12, Conjecture 1.5], that the accumulation points of the n-runner spectrum are the values of the $(n - 1)$-runner spectrum, on the range of values below $\frac{1}{3}$. Theorem 1.4 is proved in Section 9 (Theorem 9.6 and Corollary 9.7).

Above $\frac{1}{3}$ the reduction meets the five three-dimensional subtori with loneliness $\frac{2}{5}$, each of which carries infinitely many values. A version of the density lemma based on the shortest character of a subtorus, together with an exact analysis of these five subtori, reaches the next accumulation point $\frac{2}{5}$ of the three-runner spectrum. Put

$$\begin{aligned}\mathcal{R} &= \Big\{ \frac{p}{q} \in \Big[\frac{1}{3}, \frac{2}{5}\Big) : \gcd(p, q) = 1,\ 1 \leq 2q - 5p \leq 4 \Big\} \\ &= \Big\{ \frac{2j}{5j + 2} : j \geq 2 \Big\} \cup \Big\{ \frac{2j + 1}{5j + 4} : j \geq 2 \Big\}.\end{aligned}$$

Theorem 1.5 (Two-dimensional subtori below 2/5)**.** *A coordinate-free two-dimensional subtorus U of $\mathbb{T}^4$ with $\mathrm{ML}(U) < \frac{2}{5}$ has*

$$\begin{aligned}\mathrm{ML}(U) &\in \left\{\frac{s}{3s+1} : s \geq 1\right\} \cup \mathcal{R} \\ &= \left\{\frac{1}{4}, \frac{2}{7}, \frac{3}{10}, \dots\right\} \cup \left\{\frac{1}{3}, \frac{6}{17}, \frac{5}{14}, \frac{4}{11}, \frac{7}{19}, \frac{10}{27}, \frac{3}{8}, \dots\right\}.\end{aligned}$$

This set is exactly the set of three-runner values below $\frac{2}{5}$, and $\mathrm{ML}(2,3,5j) = \frac{2j}{5j+2}$, $\mathrm{ML}(1,4,5j+3) = \frac{2j+1}{5j+4}$ for $j \geq 2$. Consequently the accumulation points of the four-runner spectrum in $(0, \frac{2}{5}]$ are exactly the three-runner values there.

In the notation $D = \frac{1}{2} - \mathrm{ML}$ of [12] this gives $\mathrm{acc}(\mathcal{S}(4)) \cap [\frac{1}{10}, \frac{1}{2}) = \mathcal{S}(3) \cap [\frac{1}{10}, \frac{1}{2})$, the case $n = 4$ of [12, Conjecture 1.5] on a range that contains the two accumulation points $\frac{1}{6}$ and $\frac{1}{10}$ of $\mathcal{S}(3)$. It also determines $\mathcal{S}(3) \cap (\frac{1}{10}, \frac{1}{6}]$ exactly. Jain and Kravitz described this set up to finitely many exceptions [14, Theorem 1.4], and Corollary 9.15 shows that there are none in that interval. Finally $\frac{9}{25} \notin \mathcal{R}$. Giri and Kravitz showed that $\frac{7}{50} = \frac{1}{2} - \frac{9}{25}$ is the value of a one-dimensional closed subgroup of $\mathbb{T}^3$ but of no circle, and wrote that they believed it is not the value of a two-dimensional subtorus of $\mathbb{T}^4$ but lacked the computing power to check [12, Section 8]. Theorem 1.5 confirms this, so the spectra of two-dimensional subtori and of two-dimensional closed subgroups of $\mathbb{T}^4$ differ. Remark 9.16 explains why the full case $n = 4$ needs a new idea: the natural route through circles with a repeated speed works for every subtorus of covolume at most 20 with $\frac{1}{3} < \mathrm{ML} < \frac{1}{2}$, but fails for an explicit subtorus with $\mathrm{ML} = \frac{13}{30}$, whose value is nevertheless a three-runner value. Section 10 turns this into an infinite family. For the subtori U there, with $\mathrm{ML}(U) = \frac{1}{2} - \frac{1}{N}$, every circle in U with a repeated speed has loneliness at least N^{-3} below $\mathrm{ML}(U)$, circles in U with four distinct speeds come arbitrarily close to $\mathrm{ML}(U)$, and $\mathrm{ML}(U) = \mathrm{ML}(1, \frac{N}{2}, N-1)$ is still a three-runner value. These subtori are one phase of a family in which a weighted coalescence of two coordinates gives three-runner realizations on both sides of a phase transition.

The density lemma also works for every number of speeds. Combined with Minkowski's second theorem and a bound for the denominator of the loneliness of a two-dimensional subtorus, it gives an effective form of the reduction of Cordella.

Theorem 1.6 (Every number of speeds)**.** *Let $n \geq 3$, assume the Lonely Runner Conjecture for $n-1$ and for $n-2$ speeds, and put $\kappa_n = (n($n-*

1))$^{n-2}$. *Every set V of n positive integers with* $\gcd(V) = 1$ *satisfies*

$$\mathrm{ML}(V) > \frac{1}{n} - \frac{\kappa_n}{2|V|},$$

where $|V|$ is the Euclidean length of the vector of speeds. If moreover $\mathrm{ML}(V) < \frac{1}{n}$ *and* $|V| \geq n\kappa_n^2$*, then the circle of V lies in a two-dimensional subtorus of $(\mathbb{R}/\mathbb{Z})^n$ with loneliness exactly $\frac{1}{n}$ and covolume less than κ_n, which also contains the circle of a set of $n-1$ speeds with loneliness $\frac{1}{n}$.*

The hypotheses hold for $n \leq 15$ [2, 18, 20, 19, 1]. The second statement is the effective form of [8, Lemma 2.4], under the same hypotheses, and answers [8, Question 6.7] for every n. The first statement shows that the values below $\frac{1}{n}$ approach $\frac{1}{n}$ at rate $1/|V|$, which is sharp by Theorem 1.1(a). For counterexamples themselves sharper bounds are known [16, 1]. This is Theorem 11.1 in Section 11. The constants are too large for an exhaustive search, which is why the case $n = 4$ needs the finer analysis below.

For five speeds, Theorem 1.1 itself sharpens the general statement. Write $T_u \times \mathbb{T}$ for the product of the circle of a quadruple u with a full coordinate circle.

Theorem 1.7 (Five speeds)**.** *A coordinate-free two-dimensional subtorus of $(\mathbb{R}/\mathbb{Z})^5$ with loneliness below $\frac{2}{9}$ is, up to signed coordinate permutations, $T_{(1,2,3,4)} \times \mathbb{T}$ or $T_{(1,3,4,7)} \times \mathbb{T}$, and its loneliness is $\frac{1}{5}$. Consequently the only accumulation point of the five-speed spectrum below $\frac{2}{9}$ is $\frac{1}{5}$, and every primitive five-speed tuple V with* $\mathrm{ML}(V) < \frac{1}{5}$ *and* $|V| \geq 33{,}750$ *is, up to order, $\{a, 2a, 3a, 4a, b\}$ or $\{a, 3a, 4a, 7a, b\}$.*

The proof (Section 11.1) uses that every circle with a repeated speed in such a subtorus has at most four distinct speeds and loneliness below $\frac{2}{9}$, so by Theorem 1.1 it is the circle of $\{1, 2, 3, 4\}$ or $\{1, 3, 4, 7\}$. Together with the bounds of Section 11 this leaves a finite computation. For four speeds the second critical subtorus U^2 is not a product, so the two cases differ in kind.

Method. The main tool is a density lemma (Lemma 3.1). If a subtorus S of $\mathbb{T}^4 = (\mathbb{R}/\mathbb{Z})^4$ has a lattice of large covolume, then S is dense, in the maximum norm, in a subtorus of one higher dimension, with an explicit constant obtained from Vaaler's theorem on central sections of the cube [21] and Minkowski's convex body theorem. Since ML is 1-Lipschitz in the maximum norm, this transfers lower bounds on ML from larger subtori to smaller ones. This is the quantitative, maximum norm version of the tube argument of Giri and Kravitz [12, Lemma 4.4].

Applied to the circle T_v of a quadruple v with $\mathrm{ML}(v) < 1/4$ and $|v| \geq 512$, the lemma produces a two-dimensional subtorus $U \supseteq T_v$ with $\mathrm{ML}(U) < 5/16$. This is the effective step: it bounds the length of every near-tight quadruple that lies on no critical subtorus by 512 (Proposition 6.2), which is what [8, Question 6.7] asks for in the case of four speeds. Applied once more, it shows that every two-dimensional subtorus with $\mathrm{ML} \leq 5/16$ either has covolume less than 33 or lies in one of two explicit three-dimensional subtori with a small parameter. Both cases are enumerated exactly. The result is a list of thirty-one subtori (Proposition 5.2). Two of them are the subtori U^1 and U^2 of Jain and Kravitz with $\mathrm{ML} = 1/4$, which carry the two infinite families, and the remaining twenty-nine have $\mathrm{ML} \geq 2/7$ and cannot contain a circle of length ≥ 288 with loneliness below $1/4$. An exact exhaustive search over all primitive quadruples with $|v| < 512$ completes the classification. The formulas for the two infinite families are proved by an elementary analysis of the superlevel sets of ML on U^1 and U^2, in the spirit of [14].

All computations use exact integer or rational arithmetic. The programs, the certificate data and a Python script using only the standard library accompany the paper, see Section 12. The script recomputes the smaller computations independently and checks the certificates and output files of the large enumerations.

2. Preliminaries

Write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and $\mathbb{T}^4 = \mathbb{R}^4/\mathbb{Z}^4$. For $x \in \mathbb{T}^4$ put

$$G(x) = \min_{1 \leq i \leq 4} \|x_i\|.$$

For $p, q \in \mathbb{R}^4$ we use the maximum norm $|p|_\infty = \max_i |p_i|$ and the Euclidean norm $|p|$. Since $\|\cdot\|$ is 1-Lipschitz on $\mathbb{R}$,

$$|G(x+p) - G(x)| \leq |p|_\infty \qquad (x \in \mathbb{T}^4,\ p \in \mathbb{R}^4). \tag{3}$$

A *subtorus* of $\mathbb{T}^4$ is a closed connected subgroup, that is, the image $S = E/(E \cap \mathbb{Z}^4)$ of a rational linear subspace $E \subseteq \mathbb{R}^4$. We write $\Lambda_S = E \cap \mathbb{Z}^4$, a primitive sublattice of $\mathbb{Z}^4$ of rank $k = \dim E$, and $\mathrm{covol}(S) = \mathrm{covol}(\Lambda_S)$ for the k-dimensional volume of a fundamental parallelepiped of Λ_S in E. Put

$$\mathrm{ML}(S) = \max_{x \in S} G(x).$$

Call S *coordinate-free* if it is contained in no coordinate hyperplane $\{x_i = 0\}$, equivalently if no coordinate function vanishes identically on Λ_S. All subtori that contain a coordinate-free subtorus are coordinate-free.

For $v \in \mathbb{Z}^4$ primitive with all $v_i \neq 0$ let $T_v = \{tv : t \in \mathbb{R}\} \subseteq \mathbb{T}^4$ be its circle. Then $\Lambda_{T_v} = \mathbb{Z}v$, $\operatorname{covol}(T_v) = |v|$, and

$$\mathrm{ML}(T_v) = \max_t \min_i \|v_i t\| = \mathrm{ML}(|v_1|, |v_2|, |v_3|, |v_4|).$$

The group of signed coordinate permutations acts on $\mathbb{T}^4$ and preserves G, hence preserves ML of subtori, and it maps subtori to subtori. "Up to symmetry" always refers to this group of order 384.

Lemma 2.1 (Corner lemma)**.** *Let V be a finite set of positive integers. Then $\mathrm{ML}(V) = \min_{v \in V} \|vt\|$ for some $t = m/(u+w)$ with $u, w \in V$ and $m \in \mathbb{Z}$.*

Proof. The function $f(t) = \min_{v \in V} \|vt\|$ is continuous and piecewise linear, with pieces of slopes $\pm v$. At a maximum point t_0 with $f(t_0) > 0$, either some wave $\|vt\|$ has a downward cusp at t_0 and $f(t_0) = \|vt_0\| = 1/2$, so $2vt_0 \in \mathbb{Z}$, or two active pieces of slopes $u > 0$ and $-w < 0$ meet at t_0, so $ut_0 - a = b - wt_0$ for integers a, b and $(u+w)t_0 \in \mathbb{Z}$. The case $f \equiv 0$ is impossible for finitely many positive speeds. □

This is [15, Proposition 4.1]. It makes $\mathrm{ML}(V)$ a finite exact computation, and it shows that $\mathrm{ML}(V)$ is rational.

Lemma 2.2 (Orbit lines)**.** *Let x, y be coprime integers. A point $(X, Y) \in \mathbb{T}^2$ lies on the circle $\{(xt, yt) : t \in \mathbb{R}\}$ if and only if $yX - xY \in \mathbb{Z}$.*

Proof. Necessity is clear. If $yX - xY = k \in \mathbb{Z}$, choose $h \in \mathbb{Z}$ with $yh \equiv -k \pmod{x}$ and put $t = (X + h)/x$. Then $xt \equiv X$ and $yt = (yX + yh)/x = (k + xY + yh)/x \equiv Y \pmod 1$. □

3. The density lemma

Lemma 3.1 (Density)**.** *Let S be a k-dimensional subtorus of $\mathbb{T}^4$, $1 \leq k \leq 3$, with $A = \operatorname{covol}(S)$, and let $E = \mathbb{R}\Lambda_S$. Let $\pi : \mathbb{R}^4 \to E^\perp$ be the orthogonal projection. Then*

(a) *$\pi(\mathbb{Z}^4)$ is a lattice of rank $4 - k$ in $E^\perp$ with $\operatorname{covol}(\pi(\mathbb{Z}^4)) = 1/A$.*
(b) *there is $w \in \mathbb{Z}^4 \setminus \Lambda_S$ with $|\pi(w)|_\infty \leq A^{-1/(4-k)}$.*
(c) *for any $w \in \mathbb{Z}^4 \setminus \Lambda_S$, let S' be the $(k+1)$-dimensional subtorus with $\mathbb{R}\Lambda_{S'} = E + \mathbb{R}w$. Every point of S' lies within maximum norm distance $|\pi(w)|_\infty/2$ of a point of S. Consequently*

$$\mathrm{ML}(S) \;\geq\; \mathrm{ML}(S') - \tfrac{1}{2}|\pi(w)|_\infty.$$

In particular $\mathrm{ML}(S) \geq \mathrm{ML}(S') - \frac{1}{2}A^{-1/(4-k)}$ for a suitable $S' \supseteq S$.

Proof. (a) Since Λ_S is primitive, a basis $b_1, \dots, b_k$ of Λ_S extends to a basis $b_1, \dots, b_4$ of $\mathbb{Z}^4$. The kernel of π on $\mathbb{Z}^4$ is $\mathbb{Z}^4 \cap E = \Lambda_S$, so $\pi(\mathbb{Z}^4) \cong \mathbb{Z}^4/\Lambda_S \cong \mathbb{Z}^{4-k}$ is a lattice spanned by $\pi(b_{k+1}), \dots, \pi(b_4)$. The volume of the parallelepiped spanned by $b_1, \dots, b_4$ is 1 and it factors as the k-volume of the parallelepiped spanned by $b_1, \dots, b_k$ in E times the $(4-k)$-volume of the one spanned by $\pi(b_{k+1}), \dots, \pi(b_4)$ in $E^\perp$. Hence $1 = A \cdot \operatorname{covol}(\pi(\mathbb{Z}^4))$.

(b) Let $\rho > 0$ and $K_\rho = \{z \in E^\perp : |z|_\infty \le \rho\}$. This is the central section of the cube $[-\rho, \rho]^4$ by the $(4-k)$-dimensional subspace $E^\perp$, a compact, convex, centrally symmetric set. By Vaaler's theorem [21], every central section of the unit cube $[-\frac{1}{2}, \frac{1}{2}]^4$ by a subspace of dimension d has d-dimensional volume at least 1. Scaling by 2ρ gives $\operatorname{vol}_{4-k}(K_\rho) \ge (2\rho)^{4-k}$. Minkowski's convex body theorem (in the form valid for compact bodies, see [6, Chapter III]) states that a compact centrally symmetric convex set of volume at least $2^{4-k}\operatorname{covol}(\pi(\mathbb{Z}^4)) = 2^{4-k}/A$ contains a nonzero point of $\pi(\mathbb{Z}^4)$. For $\rho = A^{-1/(4-k)}$ we have $(2\rho)^{4-k} = 2^{4-k}/A$, so there is $w \in \mathbb{Z}^4$ with $\pi(w) \neq 0$, that is $w \notin E$, and $|\pi(w)|_\infty \le \rho$.

(c) Write $u = \pi(w)$ and $w = u + e$ with $e \in E$. Let $y \in E + \mathbb{R}w$ represent a point of S', say $y = s + bu$ with $s \in E$ and $b \in \mathbb{R}$. Choose $j \in \mathbb{Z}$ with $|b - j| \le 1/2$. Then

$$y - jw = (s - je) + (b - j)u,$$

and $y - jw \equiv y \pmod{\mathbb{Z}^4}$. The point $s - je \in E$ represents a point of S, and $(b-j)u$ has maximum norm at most $|u|_\infty/2$. By (3), $G(y) \le G(s - je) + |u|_\infty/2 \le \mathrm{ML}(S) + |u|_\infty/2$. Taking y to be a maximizer of G on S' gives the inequality. □

Remark 3.2. For $k = 3$ Vaaler's theorem is not needed: $E^\perp = \mathbb{R}m$ for a primitive $m \in \mathbb{Z}^4$, $\pi(\mathbb{Z}^4) = \mathbb{Z}\, m/|m|^2$, and the vector w with $m \cdot w = 1$ has $|\pi(w)|_\infty = |m|_\infty/|m|^2$. Note also $\operatorname{covol}(\Lambda_S) = |m|$ in this case.

4. Three-dimensional subtori

A coordinate-free three-dimensional subtorus is $W_m = \{x \in \mathbb{T}^4 : m \cdot x \in \mathbb{Z}\}$ for a primitive $m \in \mathbb{Z}^4$ with at least two nonzero entries, and $\operatorname{covol}(W_m) = |m|$. Up to symmetry we may take the entries of m nonnegative after permuting, but we keep signs since they matter for the explicit coordinates.

Lemma 4.1. *Every coordinate-free three-dimensional subtorus W satisfies* $\mathrm{ML}(W) \ge 1/3$.

Proof. Let $W = W_m$. We find $x \in [\frac{1}{3}, \frac{2}{3}]^4$ with $m \cdot x \in \mathbb{Z}$. The set $\{m \cdot x : x \in [\frac{1}{3}, \frac{2}{3}]^4\}$ is an interval of length $\frac{1}{3}\sum_i |m_i|$. If $\sum_i |m_i| \geq 3$ the interval has length at least 1 and contains an integer. Otherwise $m = \pm e_i \pm e_j$ for some $i \neq j$, and $x_i = x_j = \frac{1}{2}$ gives $m \cdot x \in \{0, \pm 1\}$. □

The two subtori

$$W^1 = \{x_1 = 2x_2\} = \{(2y, y, z, u)\},$$
$$W^2 = \{x_3 = x_1 + x_2\} = \{(a, b, a + b, u)\}$$

have ML = 1/3 exactly: the bound $\mathrm{ML}(W^1) \leq \mathrm{ML}(1, 2) = 1/3$ is clear, and $\min(\|a\|, \|b\|, \|a + b\|) \leq 1/3$ for all a, b since $\|a\|, \|b\| > 1/3$ forces $a, b \in (\frac{1}{3}, \frac{2}{3}) + \mathbb{Z}$ and then $\|a + b\| \leq \frac{1}{3}$.

Lemma 4.2. *Let W be a coordinate-free three-dimensional subtorus with $\mathrm{ML}(W) < 2/5$. Then W is, up to symmetry, W^1 or W^2.*

Proof. Let $W = W_m$. By Lemma 3.1(c) with $S = W$, $S' = \mathbb{T}^4$ and the vector w with $m \cdot w = 1$ (see the remark above), and since $G(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}) = \frac{1}{2}$,

$$\mathrm{ML}(W) \;\geq\; \frac{1}{2} - \frac{|m|_\infty}{2|m|^2}.$$

If $\mathrm{ML}(W) < 2/5$ then $|m|^2 < 5|m|_\infty$. As $|m|^2 \geq |m|_\infty^2$ this forces $|m|_\infty \leq 4$ and then $|m|^2 \leq 19$. There are exactly 21 classes of primitive $m \in \mathbb{Z}^4$ with at least two nonzero entries and $|m|^2 \leq 19$ up to symmetry. For 19 of them Table 1 exhibits a point $x \in W_m$ with $G(x) \geq 2/5$. The remaining two classes are $m = (0, 0, 1, -2)$ and $m = (0, 1, -1, -1)$, whose subtori are W^1 and W^2 up to symmetry. □

5. Two-dimensional subtori with loneliness at most 5/16

5.1. **Reduction to a finite list.** We identify W^1 with $\mathbb{T}^3$ through $(y, z, u) \mapsto (2y, y, z, u)$, and W^2 with $\mathbb{T}^3$ through $(a, b, u) \mapsto (a, b, a + b, u)$. The two-dimensional subtori of $\mathbb{T}^3$ are the kernels $U_n = \{p \in \mathbb{T}^3 : n \cdot p \in \mathbb{Z}\}$ of primitive $n \in \mathbb{Z}^3$.

Lemma 5.1. *Let U be a coordinate-free two-dimensional subtorus of $\mathbb{T}^4$ with $\mathrm{ML}(U) \leq 5/16$. Then at least one of the following holds.*

(i) $\mathrm{covol}(U) < 33$.
(ii) *Up to symmetry, $U \subseteq W^1$ or $U \subseteq W^2$, and in the coordinates of $\mathbb{T}^3$ above, $U = U_n$ with $|n| \leq 48$.*

Proof. Let $A = \mathrm{covol}(U)$. Lemma 3.1 gives a three-dimensional subtorus $W \supseteq U$ with $\mathrm{ML}(U) \geq \mathrm{ML}(W) - \frac{1}{2}A^{-1/2}$, and W is coordinate-free. If $\mathrm{ML}(W) \geq 2/5$, then $\frac{1}{2}A^{-1/2} \geq \frac{2}{5} - \frac{5}{16} = \frac{7}{80}$, so

$A \le (40/7)^2 < 33$, which is (i). Otherwise Lemma 4.2 applies, and after a symmetry $W = W^1$ or $W = W^2$.

Suppose $W = W^1$ and let $U = U_n$ in the coordinates (y, z, u), with $n \in \mathbb{Z}^3$ primitive. The function G becomes $G_1(y, z, u) = \min(\|2y\|, \|y\|, \|z\|, \|u\|)$, which satisfies $|G_1(p + \delta) - G_1(p)| \le N_1(\delta)$ with $N_1(\delta) = \max(2|\delta_1|, |\delta_2|, |\delta_3|)$. Choose $w \in \mathbb{Z}^3$ with $n \cdot w = 1$. The orthogonal projection of w onto $\mathbb{R}n$ is $n/|n|^2$, and the argument of Lemma 3.1(c), carried out in $\mathbb{T}^3$ with the norm N_1, shows that every point of $\mathbb{T}^3$ is within N_1-distance $N_1(n)/(2|n|^2)$ of U_n. The point $P = (\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ has $G_1(P) = \frac{1}{3}$, so

$$\mathrm{ML}(U_n) \;\ge\; \frac{1}{3} - \frac{N_1(n)}{2|n|^2} \;\ge\; \frac{1}{3} - \frac{|n|_\infty}{|n|^2} \;\ge\; \frac{1}{3} - \frac{1}{|n|}.$$

If $\mathrm{ML}(U_n) \le 5/16$ this gives $|n| \le 48$. For $W = W^2$ use $G_2(a, b, u) = \min(\|a\|, \|b\|, \|a + b\|, \|u\|)$, the norm $N_2(\delta) = \max(|\delta_1|, |\delta_2|, |\delta_1 + \delta_2|, |\delta_3|) \le 2|\delta|_\infty$, and the point $(\frac{1}{3}, \frac{1}{3}, \frac{1}{2})$ with $G_2 = \frac{1}{3}$. The same computation gives $|n| \le 48$. □

5.2. **The list.** We now enumerate both cases of Lemma 5.1 exactly.

For (i), every primitive rank-two sublattice of $\mathbb{Z}^4$ has a unique basis in row Hermite normal form: rows s, r with pivot columns $p < q$, $s_p > 0$, $r_q > 0$, $s_j = 0$ for $j < p$, $r_j = 0$ for $j < q$, and $0 \le s_q < r_q$. The covolume is $\left(\sum_{i<j}(s_i r_j - s_j r_i)^2\right)^{1/2}$, primitivity means that the 2×2 minors have greatest common divisor 1, and bounds on all entries follow from the bound on the minors. The enumeration produces 7,011,052 primitive coordinate-free sublattices with covolume at most 33. As an independent check of completeness, the same program counts 2,380 and 51,820 lattices of covolume at most 5 and 10, which agree with a direct count of primitive integer solutions of the Plücker relation $p_{12}p_{34} - p_{13}p_{24} + p_{14}p_{23} = 0$ of the same norms. For (ii), the primitive $n \in \mathbb{Z}^3$ with $|n| \le 48$ up to sign are listed (192,517 vectors, of which 192,514 and 192,513 give coordinate-free subtori of W^1 and W^2 respectively), a basis of $\{c \in \mathbb{Z}^3 : n \cdot c = 0\}$ is computed by integer row reduction, and the covolume of the image in $\mathbb{Z}^4$ is verified against the closed formulas $n_1^2 + 5n_2^2 + 5n_3^2$ and $2n_1^2 - 2n_1n_2 + 2n_2^2 + 3n_3^2$.

A subtorus U is discarded when some point of a grid $\{(i/Q, j/Q)\}$ in its coordinates has $G > 5/16$. The grids $Q \in \{12, 60, 420, 16, 80, 208\}$ discard all but 5,004 bases, which represent 123 subtori up to symmetry. For each of these an exact maximization (Section 12) finds a maximizer, and that point is a certificate: 92 of the 123 have a point with $G > 5/16$ and are discarded, and the remaining 31 have $\mathrm{ML} \le 5/16$.

Proposition 5.2. *Up to symmetry there are exactly* 31 *coordinate-free two-dimensional subtori* $U \subseteq \mathbb{T}^4$ *with* $\mathrm{ML}(U) \le 5/16$. *They are listed in Table 2. Exactly two of them have* $\mathrm{ML}(U) = 1/4$, *namely*

$$U^1 = \{(X, Y, 2Y, 3Y)\}, \qquad U^2 = \{(X, Y, X+Y, X+2Y)\},$$

and the other 29 *satisfy* $2/7 \le \mathrm{ML}(U) \le 5/16$ *and* $\mathrm{covol}(U)^2 \le 422$. *There is no coordinate-free two-dimensional subtorus with* $\mathrm{ML}(U) < 1/4$.

Proof. The enumeration described above is complete by Lemma 5.1, and a subtorus is discarded only on the evidence of an explicit point. For each of the 31 listed subtori the table gives a point attaining the stated value, which proves the lower bounds, in particular $\mathrm{ML} \ge 2/7$ for the 29 subtori other than U^1 and U^2. The exact values $\mathrm{ML}(U^1) = \mathrm{ML}(U^2) = 1/4$ are proved in Lemmas 8.1 and 8.3. The upper bounds $\mathrm{ML} \le 5/16$ for the other 29 come from the exact maximization and are not used in the sequel. The last sentence holds because the list contains all coordinate-free subtori with $\mathrm{ML} \le 5/16$ and the smallest value in it is $1/4$. □

Up to symmetry, U^1 and U^2 are the subtori $\langle (0,1,2,3), (1,0,0,0)\rangle$ and $\langle (1,0,1,1), (1,1,0,2)\rangle$ of [14, Section 3.3], and the last two assertions of the proposition recover the classification given there. The values occurring in Table 2 are $1/4, 2/7, 3/10, 4/13, 5/16$, that is, the values $s/(3s+1)$ of the three-runner spectrum [15, Theorem 7.2], which is consistent with the inclusion $\mathcal{S}_2(4) \subseteq \mathcal{S}_1^*(3)$ of [12].

6. Circles of length at least 512

Lemma 6.1. *Let* $T_v \subseteq U$ *with* U *a two-dimensional subtorus of covolume* A *and* $V = |v|$. *Every point of* U *lies within Euclidean distance* $A/(2V)$ *of* T_v, *and hence* $\mathrm{ML}(T_v) \ge \mathrm{ML}(U) - A/(2V)$.

Proof. In the plane $E = \mathbb{R}\Lambda_U$ the lattice Λ_U contains v as a primitive vector, so $\Lambda_U = \mathbb{Z}v + \mathbb{Z}\lambda$ for some λ, and the lines $\mathbb{R}v + k\lambda$, $k \in \mathbb{Z}$, are parallel at mutual distance A/V, because the parallelogram spanned by v and λ has area A and base V. Each of these lines maps onto T_v in $\mathbb{T}^4$. Every point of E is within $A/(2V)$ of one of the lines. The inequality follows from (3) and $|\cdot|_\infty \le |\cdot|$. □

Proposition 6.2. *Let* $v \in \mathbb{Z}^4$ *be primitive with all* $v_i \ne 0$, $\mathrm{ML}(T_v) < 1/4$ *and* $|v| \ge 512$. *Then, up to symmetry,* $T_v \subseteq U^1$ *or* $T_v \subseteq U^2$.

Proof. Put $V = |v| = \mathrm{covol}(T_v)$. Lemma 3.1 with $k = 1$ gives a two-dimensional subtorus $U \supseteq T_v$ with

$$\mathrm{ML}(U) \;\le\; \mathrm{ML}(T_v) + \tfrac{1}{2}V^{-1/3} \;<\; \tfrac{1}{4} + \tfrac{1}{16} = \tfrac{5}{16}.$$

U is coordinate-free, so by Proposition 5.2 it is one of the 31 subtori up to symmetry. If U is one of the 29 subtori with $\mathrm{ML}(U) \geq 2/7$, Lemma 6.1 gives $\frac{1}{4} > \mathrm{ML}(T_v) \geq \frac{2}{7} - \frac{A}{2V}$, hence $V < 14A \leq 14\sqrt{422} < 288$, a contradiction. So U is U^1 or U^2 up to symmetry. □

Remark 6.3. Nothing in Lemma 3.1 is special to dimension four. In $(\mathbb{R}/\mathbb{Z})^n$ the same argument gives a vector w with $|\pi(w)|_\infty \leq \mathrm{covol}(S)^{-1/(n-k)}$, so a circle of length V is $\frac{1}{2}V^{-1/(n-1)}$-dense in a two-dimensional subtorus. Section 11 uses the n-dimensional form of the lemma to make the reduction of near-tight tuples to critical subtori in [8, Lemma 2.4] effective for every n.

Lemma 6.4. *Let $v \in \mathbb{Z}^4$ be primitive with all $v_i \neq 0$. If $T_v \subseteq U^1$ up to symmetry, then $\{|v_i|\} = \{a, b, 2b, 3b\}$ for integers $a, b \geq 1$. If $T_v \subseteq U^2$ up to symmetry, then $\{|v_i|\} = \{x, y, x+y, x+2y\}$ for integers $x, y \geq 1$.*

Proof. $T_v \subseteq U$ means $v \in \Lambda_U$. The lattices are $\Lambda_{U^1} = \mathbb{Z}(1,0,0,0) + \mathbb{Z}(0,1,2,3)$ and $\Lambda_{U^2} = \mathbb{Z}(1,0,1,1) + \mathbb{Z}(0,1,1,2)$ (their Plücker coordinates have greatest common divisor 1, so these lattices are primitive). A signed permutation of v is therefore $(a, b, 2b, 3b)$, respectively $(x, y, x+y, x+2y)$, for integers a, b, respectively x, y, and the multiset of absolute values is $\{|a|, |b|, 2|b|, 3|b|\}$, respectively $\{|x|, |y|, |x+y|, |x+2y|\}$. In the first case replace a, b by their absolute values. In the second case, if x and y have opposite signs, say $x > 0 > y$ and $y' = -y$, then the multiset is $\{x, y', |x-y'|, |x-2y'|\}$. If $x > 2y'$ it equals $V(x-2y', y')$, if $y' < x < 2y'$ it equals $V(2y'-x, x-y')$, and if $x < y'$ it equals $V(x, y'-x)$. The cases $x = y'$ and $x = 2y'$ produce a zero coordinate and are excluded. □

7. The exhaustive search and the proof of Theorem 1.1

By Proposition 6.2 and Lemma 6.4, every primitive quadruple v with $\mathrm{ML}(v) < 1/4$ is of the form $\{a, b, 2b, 3b\}$ or $\{x, y, x+y, x+2y\}$, or satisfies $|v| < 512$. The two families are analyzed in Section 8. For the remaining case we enumerate all 800,816,099 primitive quadruples $v_1 < v_2 < v_3 < v_4$ with $v_1^2 + v_2^2 + v_3^2 + v_4^2 < 512^2$ and decide for each whether $\mathrm{ML}(v) < 1/4$. A quadruple is accepted as soon as some time $t = j/Q$ from a fixed list of 727 rational times with $Q \leq 512$ has $\|v_i t\| \geq \frac{1}{4}$ for all i. The 16,649 quadruples that pass no such test are handled by the exact corner computation of Lemma 2.1. Exactly 14,032 quadruples have $\mathrm{ML} < 1/4$. All of them belong to one of the two families, with the single exception

$$v = (1, 3, 4, 14), \qquad \mathrm{ML}(1, 3, 4, 14) = \tfrac{4}{17},$$

attained at $t = 7/17$, where $(\|7/17\|, \|21/17\|, \|28/17\|, \|98/17\|) = (7, 4, 6, 4)/17$. This quadruple was first observed by Cordella [8, Section 6.3], whose search covered all quadruples with speeds at most 200. Restricting our output to that range reproduces his counts exactly: 4,125 primitive quadruples with ML $< 1/4$, of which 3,836 have $k = 1$ and 289 have $k = 2$ in the notation ML $= s/(4s + k)$, with 50 on U^1 and 4,075 on U^2 (one quadruple on both) and one on neither. That $(1, 3, 4, 14)$ is in neither family is immediate: it contains no triple $\{b, 2b, 3b\}$, and the two smallest elements of $V(x, y)$ are x and y, so $V(x, y) = \{1, 3, 4, 14\}$ would force $\{x, y\} = \{1, 3\}$, but $V(1, 3) = \{1, 3, 4, 7\}$ and $V(3, 1) = \{1, 3, 4, 5\}$. The quadruple lies on the subtorus $\{(X, Y, X + Y, 2X + 4Y)\}$, which appears in Table 2 with ML $= 2/7$ and covolume $\sqrt{27}$.

Proof of Theorem 1.1. Let V be a primitive quadruple with $\mathrm{ML}(V) < 1/4$. By the discussion above, V is $\{a, b, 2b, 3b\}$, or $V(x, y)$, or $\{1, 3, 4, 14\}$. Theorem 8.2 shows that $\{a, b, 2b, 3b\}$ has ML $< 1/4$ only for $b = 1$ and $4 \mid a$, with the value in (a). Theorem 8.4 shows that $V(x, y)$ has ML $< 1/4$ exactly under the congruence conditions in (b), with the stated value, and gives $\mathrm{ML}(V(x, y)) = 1/4$ otherwise. Conversely each listed quadruple has the stated value below $1/4$. For the overlap statement, note that the two smallest elements of $V(x, y)$ are x and y, so $\{1, 2, 3, 4k\} = V(x, y)$ forces $\{x, y\} = \{1, 2\}$, and $V(1, 2) = \{1, 2, 3, 5\}$ is not of the form $\{1, 2, 3, 4k\}$ while $V(2, 1) = \{1, 2, 3, 4\}$ is. The sporadic quadruple belongs to neither family by Section 7. □

8. The two infinite families

For a circle $T \subseteq U$ in a two-dimensional subtorus, the value $\mathrm{ML}(T)$ is the largest d such that T meets the superlevel set $S_d = \{x \in U : G(x) \geq d\}$. We describe S_d explicitly on U^1 and U^2 and use Lemma 2.2.

8.1. **The family $\{a, b, 2b, 3b\}$.** Identify U^1 with $\mathbb{T}^2$ through $(X, Y) \mapsto (X, Y, 2Y, 3Y)$, so that $G = \min(\|X\|, \|Y\|, \|2Y\|, \|3Y\|)$, and the circle of (a, b) is $\{(at, bt)\}$.

Lemma 8.1. *For $\frac{1}{5} < d \leq \frac{1}{4}$,*

$$S_d = \big[d, 1 - d\big] \times \Big(\big[d, \tfrac{1-d}{3}\big] \cup \big[\tfrac{2+d}{3}, 1 - d\big]\Big) \subseteq [0, 1)^2.$$

In particular $\mathrm{ML}(U^1) = 1/4$*, attained exactly on the two segments $Y \in \{\frac{1}{4}, \frac{3}{4}\}$, $X \in [\frac{1}{4}, \frac{3}{4}]$.*

Proof. The condition on X is $\|X\| \geq d$. For $Y \in [0, \frac{1}{2}]$ the three conditions $\|Y\|, \|2Y\|, \|3Y\| \geq d$ read $Y \in [d, 1-d]$, $Y \in [\frac{d}{2}, \frac{1-d}{2}]$, and $Y \in [\frac{d}{3}, \frac{1-d}{3}] \cup [\frac{1+d}{3}, \frac{2-d}{3}]$. Their intersection is $[d, \frac{1-d}{3}] \cup [\frac{1+d}{3}, \frac{1-d}{2}]$, and the second interval is empty exactly when $d > \frac{1}{5}$. The first is nonempty exactly when $d \leq \frac{1}{4}$. The symmetry $Y \mapsto 1 - Y$ gives the other piece. □

Theorem 8.2. *Let $a, b \geq 1$ be coprime with $a \notin \{b, 2b, 3b\}$. Then $\mathrm{ML}(a, b, 2b, 3b) = 1/4$ unless $b = 1$ and $4 \mid a$, in which case $\mathrm{ML}(1, 2, 3, a) = a/(4a + 4)$.*

Proof. Since U^1 contains the circle, $\mathrm{ML}(a, b, 2b, 3b) \leq \mathrm{ML}(U^1) = \frac{1}{4}$. By Lemma 2.2 the circle is $\{bX - aY \in \mathbb{Z}\}$. Its image under $\phi(X, Y) = bX - aY$ on the segment $Y = \frac{1}{4}$, $X \in [\frac{1}{4}, \frac{3}{4}]$ is an interval of length $\frac{b}{2}$. If $b \geq 2$ it contains an integer, so $\mathrm{ML} = \frac{1}{4}$. Let $b = 1$. The interval is $[\frac{1-a}{4}, \frac{3-a}{4}]$, which contains an integer exactly when $4 \nmid a$. Let $a = 4k$. On the rectangle $R_d = [d, 1-d] \times [d, \frac{1-d}{3}]$ the image of ϕ is $[\, d - \frac{4k(1-d)}{3},\ 1 - d - 4kd \,]$. As d decreases from $\frac{1}{4}$, the upper end increases from $\frac{3}{4} - k$ and reaches the integer $1 - k$ at $d = \frac{k}{4k+1}$, while the lower end decreases from $\frac{1}{4} - k$ and reaches the integer $-k$ at $d = \frac{k}{4k+3} < \frac{k}{4k+1}$. The other rectangle behaves in the same way by the symmetry $(X, Y) \mapsto (1 - X, 1 - Y)$. Since $\frac{k}{4k+1} \geq \frac{1}{5}$ for $k \geq 1$, Lemma 8.1 applies for all $d \in (\frac{1}{5}, \frac{1}{4}]$ and the first contact of the circle with S_d occurs at $d = \frac{k}{4k+1}$. If $k \geq 2$ this is above $\frac{1}{5}$ and we are done. If $k = 1$ the circle misses S_d for $d > \frac{1}{5}$, so $\mathrm{ML} \leq \frac{1}{5}$, and the contact at $d = \frac{1}{5}$ lies in the rectangle, which is part of $S_{1/5}$, so $\mathrm{ML} = \frac{1}{5}$. Finally $\frac{k}{4k+1} = \frac{a}{4a+4}$. □

The values $k/(4k + 1)$ are Kravitz's family $\mathrm{ML}(1, 2, \ldots, n-1, ns) = s/(ns + 1)$ [15, Theorem 5.1], here with $n = 4$.

8.2. **The family $V(x, y)$.** Identify U^2 with $\mathbb{T}^2$ through $(X, Y) \mapsto (X, Y, X + Y, X + 2Y)$, so that

$$G(X, Y) = \min\big(\|X\|, \|Y\|, \|X + Y\|, \|X + 2Y\|\big),$$

and the circle of (x, y) is $\{yX - xY \in \mathbb{Z}\}$ by Lemma 2.2.

Lemma 8.3. *For $0 < d \le \frac{1}{4}$, the set $S_d \cap [0,1)^2$ is the disjoint union of the four triangles*

$$\begin{aligned}
\Delta_1(d) &= \{X \ge d,\ Y \ge d,\ X + 2Y \le 1 - d\},\\
\Delta_2(d) &= (1,1) - \Delta_1(d),\\
\Delta_3(d) &= \{X \ge d,\ X + Y \le 1 - d,\ X + 2Y \ge 1 + d\},\\
\Delta_4(d) &= (1,1) - \Delta_3(d).
\end{aligned}$$

Writing $d = \frac{1}{4} - \varepsilon$, these are $c_j + \varepsilon Q_j$ with centres and vertex sets

$$\begin{aligned}
c_1 &= (\tfrac{1}{4}, \tfrac{1}{4}), & Q_1 &= \operatorname{conv}\{(-1,-1),(-1,1),(3,-1)\},\\
c_3 &= (\tfrac{1}{4}, \tfrac{1}{2}), & Q_3 &= \operatorname{conv}\{(-1,0),(-1,2),(3,-2)\},
\end{aligned}$$

and $c_2 = -c_1$, $Q_2 = -Q_1$, $c_4 = -c_3$, $Q_4 = -Q_3$. In particular $\mathrm{ML}(U^2) = \frac{1}{4}$, attained exactly at c_1, c_2, c_3, c_4.

Proof. For $(X,Y) \in [0,1)^2$ we have $X, Y \in [0,1)$, $X + Y \in [0,2)$ and $X + 2Y \in [0,3)$. The condition $G \ge d$ means $d \le X \le 1 - d$, $d \le Y \le 1 - d$, $k_3 + d \le X + Y \le k_3 + 1 - d$ for some $k_3 \in \{0,1\}$ and $k_4 + d \le X + 2Y \le k_4 + 1 - d$ for some $k_4 \in \{0,1,2\}$. The pair $(k_3, k_4) = (0,2)$ is impossible, since $X + Y \le 1 - d$ and $X + 2Y \ge 2 + d$ give $Y \ge 1 + 2d$. The pair $(1,0)$ is impossible, since $X + Y \ge 1 + d$ and $X + 2Y \le 1 - d$ give $Y \le -2d$. For $(0,0)$ the conditions reduce to $\Delta_1(d)$: the inequalities $X \ge d$, $Y \ge d$, $X + 2Y \le 1 - d$ imply $X + Y \le 1 - 2d$, $X \le 1 - 3d$, $Y \le \frac{1-2d}{2}$, $X + Y \ge 2d$ and $X + 2Y \ge 3d$, which are all the remaining constraints. For $(0,1)$ the conditions reduce to $\Delta_3(d)$: the inequalities $X \ge d$, $X + Y \le 1 - d$, $X + 2Y \ge 1 + d$ imply $Y \ge 2d$, $Y \le 1 - 2d$, $X \le 1 - 3d$, $X + Y \ge 3d$ and $X + 2Y \le 2 - 3d$. The pairs $(1,2)$ and $(1,1)$ are the images of $(0,0)$ and $(0,1)$ under $(X,Y) \mapsto (1 - X, 1 - Y)$, which preserves G. The vertices of $\Delta_1(d)$ are (d,d), $(d, \frac{1-2d}{2})$ and $(1 - 3d, d)$, those of $\Delta_3(d)$ are $(d, \frac{1}{2})$, $(d, 1 - 2d)$ and $(1 - 3d, 2d)$, and substituting $d = \frac{1}{4} - \varepsilon$ gives the stated form. The triangles are nonempty exactly for $d \le \frac{1}{4}$ and reduce to the centres at $d = \frac{1}{4}$. □

Theorem 8.4. *Let $x, y \ge 1$ be coprime with $x \ne y$, and let $r = (y - x) \bmod 4$, $r' = (y - 2x) \bmod 4$. Then $\mathrm{ML}(V(x,y)) = \frac{1}{4}$ if $r = 0$ or $r' = 0$, and otherwise*

(4)

$$\mathrm{ML}(V(x,y)) = \frac{1}{4} - \min\Big\{\frac{r}{4(x+y)}, \frac{4-r}{4(x+3y)}, \frac{r'}{4(2x+y)}, \frac{4-r'}{4(2x+3y)}\Big\}.$$

Proof. Let $\phi(X,Y) = yX - xY$. By Lemma 8.3, the circle meets S_d, $d = \frac{1}{4} - \varepsilon$, if and only if some integer lies in $\phi(c_j + \varepsilon Q_j)$ for some j.

The image of a triangle is the interval between the smallest and largest values at its vertices. For Q_1 the vertex values of ϕ are $x-y$, $-x-y$ and $3y+x$, so $\phi(c_1+\varepsilon Q_1) = [\,c-\varepsilon(x+y),\ c+\varepsilon(x+3y)\,]$ with $c = \phi(c_1) = \frac{y-x}{4}$, where we used $x, y > 0$. The nearest integers to c are $c - \frac{r}{4}$ and $c + \frac{4-r}{4}$ when $r \neq 0$, and c itself when $r = 0$. So the circle meets $\Delta_1(d)$ exactly when $r = 0$ or $\varepsilon \geq \min\{\frac{r}{4(x+y)}, \frac{4-r}{4(x+3y)}\}$. Since $\phi(-p) = -\phi(p)$, the triangle $\Delta_2(d)$ gives the same condition. For Q_3 the vertex values are $-y$, $-y-2x$ and $3y+2x$, the centre value is $\phi(c_3) = \frac{y-2x}{4}$, and the condition is $r' = 0$ or $\varepsilon \geq \min\{\frac{r'}{4(2x+y)}, \frac{4-r'}{4(2x+3y)}\}$, again shared by $\Delta_4(d)$. Thus $\mathrm{ML}(V(x,y)) = \frac{1}{4} - \varepsilon^*$ with ε^* the smallest $\varepsilon \geq 0$ meeting one of the conditions, which is the statement. The description of S_d is valid for all $d \in (0, \frac{1}{4}]$, and $\varepsilon^* < \frac{1}{4}$ because $\mathrm{ML}(V(x,y)) > 0$, so no further case distinction is needed. □

Lemma 8.5. *In the situation of Theorem 8.4 with $r, r' \neq 0$, the minimum in* (4) *equals $\frac{1}{4(2m+1)}$ with $m = m(x,y) = \max(m_1, m_3)$ as in* (2)*, where m_1, m_3 are integers. Consequently* $\mathrm{ML}(V(x,y)) = m/(4m+2)$.

Proof. Compare the two terms coming from c_1. If $r = 1$, then $\frac{1}{4(x+y)} \leq \frac{3}{4(x+3y)}$ since $x + 3y \leq 3x + 3y$. If $r = 3$, then $\frac{1}{4(x+3y)} \leq \frac{3}{4(x+y)}$. If $r = 2$, then $\frac{2}{4(x+3y)} \leq \frac{2}{4(x+y)}$. So the smaller of the two is $\frac{1}{4(2m_1+1)}$ with $2m_1 + 1 = x + y$, $\frac{x+3y}{2}$, $x + 3y$ for $r = 1, 2, 3$. These are odd integers: $y \equiv x + 1 \pmod 4$ gives $x + y \equiv 2x + 1 \pmod 4$, odd. Next, $y \equiv x + 2$ gives $x + 3y \equiv 4x + 6 \equiv 2 \pmod 4$, so $\frac{x+3y}{2}$ is odd. Finally $y \equiv x+3$ gives $x+3y \equiv 4x+9$, odd. The same comparison for c_3 yields $2m_3 + 1 = 2x + y$, $\frac{2x+3y}{2}$, $2x + 3y$ for $r' = 1, 2, 3$, odd by $y \equiv 2x + 1$, $y \equiv 2x + 2$ (then $2x + 3y \equiv 8x + 6 \equiv 2 \pmod 4$) and $y \equiv 2x + 3$. The minimum of the four terms is $\frac{1}{4(2m+1)}$ with $m = \max(m_1, m_3)$, and $\frac{1}{4} - \frac{1}{4(2m+1)} = \frac{m}{4m+2}$. □

Proof of Theorem 1.2. By Theorem 1.1, every value below $\frac{1}{4}$ is $k/(4k+1) = 2k/(4 \cdot 2k + 2)$, or $m(x,y)/(4m(x,y)+2)$, or $4/17 = 8/34$. All are of the form $m/(4m+2)$ with $m \geq 1$. We show $m \geq 2$, $m \neq 3$, and that every other m occurs.

$m \geq 2$. The value $1/6$ would contradict $\mathrm{ML} \geq \frac{1}{5}$, which is the Lonely Runner Conjecture for four speeds [10, 4]. Directly: $m_1 = 1$ requires $r = 1$ and $x+y = 3$, or $r = 2$ and $x+3y = 6$, so $(x,y) \in \{(1,2), (3,1)\}$. For $(1,2)$ one has $r' = 0$, and $(3,1)$ has $r' = 3$ and $m_3 = 4$. And $m_3 = 1$ requires $r' = 1$ and $2x + y = 3$, which forces $x = y = 1$, excluded, or $r' \in \{2, 3\}$ and $2x + 3y \in \{6, 3\}$, which has no solution with $x, y \geq 1$ coprime.

$m \neq 3$. Suppose $m(x,y) = 3$. If $m_1 = 3$ then either $r = 1$ and $x + y = 7$ with $y \equiv x + 1 \pmod 4$, so $(x,y) \in \{(1,6),(3,4),(5,2)\}$, or $r = 3$ and $x+3y = 7$ with $y \equiv x+3$, which has no solution with $x, y \geq 1$, or $r = 2$ and $x + 3y = 14$ with $y \equiv x + 2$, so $(x,y) \in \{(5,3),(11,1)\}$ after discarding $(2,4)$ and $(8,2)$, which are not coprime. For $(1,6)$ and $(5,2)$ one has $r' = 0$, while $(3,4)$, $(5,3)$, $(11,1)$ give $m_3 = 4, 6, 12$, all larger than 3. If $m_3 = 3$ then either $r' = 1$ and $2x + y = 7$ with $y \equiv 2x + 1$, which fails for $(1,5),(2,3),(3,1)$, or $r' = 3$ and $2x + 3y = 7$ with $y \equiv 2x + 3$, which fails for $(2,1)$, or $r' = 2$ and $2x + 3y = 14$ with $y \equiv 2x + 2$, which leaves $(1,4)$ after discarding $(4,2)$, and then $r = 3$ gives $m_1 = 6 > 3$. So $m = 3$ never occurs.

Attainment. Even $m = 2k$ occurs for $\{1, 2, 3, 4k\}$. For $(x,y) = (1,7)$ one has $r = 2$, $m_1 = 5$, $r' = 1$, $m_3 = 4$, so $\mathrm{ML}(1,7,8,15) = 5/22$. For $(x,y) = (4s+3, 8)$, $s \geq 0$, one has $r = 1$, $m_1 = 2s + 5$, $r' = 2$, $m_3 = 2s + 7$, so $\mathrm{ML}(V(4s+3,8)) = (2s+7)/(8s+30)$, which is the family of Fan and Sun and covers every odd $m \geq 7$. □

Proof of Corollary 1.3. (i) $m/(4m+2)$ equals $s/(4s+1)$ with $s = m/2$ if m is even, and $s/(4s+2)$ with $s = m$ if m is odd. (ii) ML $= \frac{1}{5}$ means $m = 2$. In family (a) this is $k = 1$. In family (b), $m_1 = 2$ requires $r = 1$ and $x + y = 5$, giving $(2,3)$ and $(4,1)$, or $r = 3$ and $x + 3y = 5$, giving $(2,1)$, or $r = 2$ and $x + 3y = 10$, giving $(1,3)$ and $(7,1)$. Similarly $m_3 = 2$ requires $r' = 1$ and $2x+y = 5$, giving $(1,3)$ and $(2,1)$, or $r' = 3$ and $2x + 3y = 5$, giving $(1,1)$, excluded, or $r' = 2$ and $2x + 3y = 10$, giving $(2,2)$, excluded. Among these, $m(x,y) = 2$ holds exactly for $(2,1)$ and $(1,3)$, since $(2,3)$, $(4,1)$, $(7,1)$ have $m_3 = 6, 4, 8$. These give $\{1,2,3,4\}$ and $\{1,3,4,7\}$. (iii) The next value is $2/9$. □

9. Two-dimensional subtori below one third

The list of Section 5 stops at $5/16$ because the circle argument needs no more. The same tools determine every coordinate-free two-dimensional subtorus with loneliness below $1/3$, and the answer has a clean form: the values are exactly the three-runner values $s/(3s+1)$ of [15, Theorems 5.1 and 7.2], and the subtori are products with a three-runner circle or members of one explicit two-parameter family. Write

$$F(n_1, n_2) = \{(a,\ b,\ a+b,\ n_1 a + n_2 b) : (a,b) \in \mathbb{T}^2\},$$

for $(n_1, n_2) \in \mathbb{Z}^2 \setminus \{0\}$, a coordinate-free two-dimensional subtorus of W^2, and for a triple (a,b,c) of positive integers write $T_{(a,b,c)} \times \mathbb{T} = \{(at, bt, ct, u)\}$, a product subtorus with $\mathrm{ML}(T_{(a,b,c)} \times \mathbb{T}) = \mathrm{ML}(a,b,c)$.

Note that $F(1,2) = U^2$, that $(1,3,4,14)$ lies on $F(2,4)$, and that U^1 is the product $T_{(1,2,3)} \times \mathbb{T}$ up to symmetry.

Lemma 9.1. *Let U be a coordinate-free two-dimensional subtorus with $\mathrm{ML}(U) < 1/3$. Then either $\mathrm{covol}(U) < 56.25$, or U lies, up to symmetry, in W^1 or in W^2.*

Proof. As in the proof of Lemma 5.1: Lemma 3.1 gives $W \supsetneq U$ with $\mathrm{ML}(U) \geq \mathrm{ML}(W) - \frac{1}{2}A^{-1/2}$, and if W is not special then $\frac{1}{2}A^{-1/2} > \frac{2}{5} - \frac{1}{3} = \frac{1}{15}$, so $A < (15/2)^2$. □

Lemma 9.2 (Subtori of W^1)**.** *Let $U = U_n \subseteq W^1$ be coordinate-free, with $n = (n_1, n_2, n_3) \in \mathbb{Z}^3$ primitive in the coordinates (y, z, u) of $W^1 = \{(2y, y, z, u)\}$. If $(n_2, n_3) \in \{(\pm 1, 0), (0, \pm 1)\}$ then U is the product $T_{(1,2,|n_1|)} \times \mathbb{T}$ up to symmetry, so $\mathrm{ML}(U) = \mathrm{ML}(1, 2, |n_1|)$, which equals $s/(3s+1)$ if $|n_1| = 3s$ and $1/3$ otherwise. In every other case $\mathrm{ML}(U) = 1/3$.*

Proof. Always $\mathrm{ML}(U) \leq \mathrm{ML}(W^1) = 1/3$. The case $(n_2, n_3) = (0, 0)$ forces $n = \pm e_1$ and $y \in \mathbb{Z}$, which is not coordinate-free. If $(n_2, n_3) = (\pm 1, 0)$ then $z = \mp n_1 y$ on U and u is free, so $U = \{(2y, y, \mp n_1 y, u)\}$ is the stated product ($n_1 \neq 0$ since U is coordinate-free). The circle $(2y, y, n_1 y)$ has speeds $\{1, 2, |n_1|\}$, and $\mathrm{ML}(1, 2, k) = 1/3$ for $3 \nmid k$ by the time $t = 1/3$ and the bound $\mathrm{ML}(1, 2) = 1/3$, while $\mathrm{ML}(1, 2, 3s) = s/(3s + 1)$ by [15, Theorem 5.1]. The case $(0, \pm 1)$ is the same with z, u exchanged. In the remaining cases we exhibit a point of U with $G = 1/3$: take $y = 1/3$ and $(z, u) \in [\frac{1}{3}, \frac{2}{3}]^2$ with $n_2 z + n_3 u \equiv -n_1/3 \pmod 1$. The set $\{n_2 z + n_3 u : z, u \in [\frac{1}{3}, \frac{2}{3}]\}$ is an interval of length $(|n_2| + |n_3|)/3$. If $|n_2| + |n_3| \geq 3$ it contains every residue class. If $(n_2, n_3) = (\pm 1, \pm 1)$ or $(\pm 2, 0)$ or $(0, \pm 2)$ it is, up to sign, $[\frac{2}{3}, \frac{4}{3}]$ or $[-\frac{1}{3}, \frac{1}{3}]$, and both contain a representative of each of the classes $0, \frac{1}{3}, \frac{2}{3} \pmod 1$, which are the only possible values of $-n_1/3$. □

Lemma 9.3 (Sum triangle)**.** *For $0 < d \leq \frac{1}{3}$,*

$$\{(a, b) \in [0, 1)^2 : \|a\|, \|b\|, \|a + b\| \geq d\} = \Delta(d) \cup \big((1, 1) - \Delta(d)\big),$$

where $\Delta(d) = \{a \geq d,\ b \geq d,\ a + b \leq 1 - d\}$. With $d = \frac{1}{3} - \varepsilon$,

$$\Delta(d) = \left(\tfrac{1}{3}, \tfrac{1}{3}\right) + \varepsilon \,\mathrm{conv}\{(-1, -1), (-1, 2), (2, -1)\}.$$

Proof. The constraints are $a, b \in [d, 1 - d]$ and $a + b \in [d, 1 - d] \cup [1 + d, 2 - d]$. In the first case $a + b \leq 1 - d$ together with $a, b \geq d$ already implies $a, b \leq 1 - 2d$ and $a + b \geq 2d$, which gives $\Delta(d)$. The second case is its image under $(a, b) \mapsto (1 - a, 1 - b)$. The vertices of $\Delta(d)$ are (d, d), $(d, 1 - 2d)$ and $(1 - 2d, d)$. □

Lemma 9.4 (Subtori of W^2)**.** *Let $U = U_n \subseteq W^2$ be coordinate-free, with $n = (n_1, n_2, n_3)$ primitive in the coordinates (a, b, u) of $W^2 = \{(a, b, a + b, u)\}$.*

(i) *If $|n_3| \geq 2$ then $\mathrm{ML}(U) = 1/3$.*
(ii) *If $n_3 = 0$ then $U = T_{(|n_2|,|n_1|,|n_1-n_2|)} \times \mathbb{T}$ up to symmetry, and $\mathrm{ML}(U)$ is the loneliness of that triple.*
(iii) *If $|n_3| = 1$ then $\mathrm{ML}(U) = \mathrm{ML}(F(n_1, n_2))$, and*

$$\mathrm{ML}(F(n_1,n_2)) = \begin{cases} \dfrac{1}{3}, & n_1 + n_2 \not\equiv 0 \pmod 3, \\ \dfrac{M}{3(M+1)} = \dfrac{s}{3s+1}, & n_1 + n_2 \equiv 0 \pmod 3, \end{cases}$$

where in the second case $M = 3s = \max\big(|n_1 + n_2|, |2n_1 - n_2|, |2n_2 - n_1|\big)$.

Proof. Always $\mathrm{ML}(U) \leq \mathrm{ML}(W^2) = \frac{1}{3}$, and $G_2(\frac{1}{3}, \frac{1}{3}, u) = \frac{1}{3}$ for every $u \in [\frac{1}{3}, \frac{2}{3}]$.

(i) We need $u \in [\frac{1}{3}, \frac{2}{3}]$ with $n_3 u \equiv -(n_1 + n_2)/3 \pmod 1$. The set $\{n_3 u\}$ is an interval of length $|n_3|/3$, which contains every class if $|n_3| \geq 3$, and for $|n_3| = 2$ it is $\pm[\frac{2}{3}, \frac{4}{3}]$, which meets each of the classes $0, \frac{1}{3}, \frac{2}{3}$.

(ii) Since n is primitive, $\gcd(n_1, n_2) = 1$, and by Lemma 2.2 the set $\{(a, b) : n_1 a + n_2 b \in \mathbb{Z}\}$ is the circle $\{(n_2 t, -n_1 t)\}$. Hence $U = \{(n_2 t, -n_1 t, (n_2 - n_1)t, u)\}$, with $n_1, n_2 \neq 0$ and $n_1 \neq n_2$ because U is coordinate-free.

(iii) On U the coordinate u equals $\mp(n_1 a + n_2 b)$, and G is invariant under changing the sign of a coordinate, so $\mathrm{ML}(U) = \mathrm{ML}(F(n_1, n_2))$. Put $\ell(a, b) = n_1 a + n_2 b$ and $d = \frac{1}{3} - \varepsilon$. By Lemma 9.3, $F(n_1, n_2)$ has a point with $G \geq d$ exactly when some point of $\Delta(d) \cup ((1, 1) - \Delta(d))$ has $\|\ell\| \geq d$. The image of $\Delta(d)$ under ℓ is the interval $[\,\ell(c) - \varepsilon L,\ \ell(c) + \varepsilon H\,]$ with $c = (\frac{1}{3}, \frac{1}{3})$, $\ell(c) = (n_1 + n_2)/3$, and $-L$, H the smallest and largest of the vertex values $-(n_1 + n_2)$, $-n_1 + 2n_2$, $2n_1 - n_2$ of ℓ on $\mathrm{conv}\{(-1, -1), (-1, 2), (2, -1)\}$. The image of $(1, 1) - \Delta(d)$ is the reflection of this interval in $\ell(1, 1)/2$, which preserves distances to $\mathbb{Z}$. If $n_1 + n_2 \not\equiv 0 \pmod 3$ then $\|\ell(c)\| = \frac{1}{3}$ and $\varepsilon = 0$ works. If $n_1 + n_2 \equiv 0 \pmod 3$ then $\ell(c) \in \mathbb{Z}$, the image interval contains a point at distance at least $\frac{1}{3} - \varepsilon$ from $\mathbb{Z}$ exactly when $\varepsilon \max(L, H) \geq \frac{1}{3} - \varepsilon$, and $\max(L, H) = M$. The first contact is at $\varepsilon = 1/(3(M + 1))$, which lies in $(0, \frac{1}{6}]$, so Lemma 9.3 applies throughout and $\mathrm{ML}(F(n_1, n_2)) = \frac{1}{3} - \frac{1}{3(M+1)}$. Finally M is one of $|n_1 + n_2|$, $|3n_1 - (n_1 + n_2)|$, $|3n_2 - (n_1 + n_2)|$, all divisible by 3. □

Proposition 9.5. *There are* 63,906,892 *coordinate-free primitive rank-two sublattices of* $\mathbb{Z}^4$ *with covolume at most* 57. *Of these,* 3,267,840 *are contained in a signed permutation image of* W^1 *or* W^2*, and each of the remaining* 60,639,052 *has a point* $(i/Q, j/Q)$*,* $Q \in \{3, 12, 60, 420, 16, 80, 208\}$*, at which* $G \geq 1/3$*. Consequently every coordinate-free two-dimensional subtorus with* $\mathrm{ML} < 1/3$ *lies in* W^1 *or* W^2 *up to symmetry.*

Proof. The enumeration is the one of Section 5 with the larger bound (the count 63,906,892 is again checked against the Plücker count for small covolumes). Containment in a special subtorus is decided by testing the characters $\pm e_i \pm 2e_j$ and $\pm e_i \pm e_j \pm e_k$ (80 vectors, 40 up to sign) against both basis vectors. The last sentence follows from Lemma 9.1. □

Theorem 9.6. *Let* U *be a coordinate-free two-dimensional subtorus of* $\mathbb{T}^4$*. Then* $\mathrm{ML}(U) < 1/3$ *if and only if, up to symmetry,* U *is a product* $T_{(a,b,c)} \times \mathbb{T}$ *with* $\mathrm{ML}(a, b, c) < 1/3$*, or* $U = F(n_1, n_2)$ *with* $n_1 + n_2 \equiv 0 \pmod 3$*. In both cases* $\mathrm{ML}(U) = s/(3s + 1)$ *for an integer* $s \geq 1$*, namely the* s *of* [15, Theorem 7.2] *for the triple (every* $s \geq 1$ *occurs, by* [15, Theorem 5.1]*), and* $s = \frac{1}{3}\max(|n_1 + n_2|, |2n_1 - n_2|, |2n_2 - n_1|)$ *for* $F(n_1, n_2)$*. Every value* $s/(3s + 1)$*,* $s \geq 1$*, occurs, for instance on* $F(s, 2s)$*.*

Proof. The subtori listed have the stated values by Lemma 9.4 and by [15, Theorems 5.1 and 7.2], which together say that the values of ML on triples below 1/3 are exactly the numbers $s/(3s+1)$, each attained by $(1, 2, 3s)$. Conversely, if $\mathrm{ML}(U) < 1/3$, Proposition 9.5 places U in W^1 or W^2 up to symmetry, and Lemmas 9.2 and 9.4 leave only products with $\mathrm{ML} < 1/3$ and $F(n_1, n_2)$ with $3 \mid n_1 + n_2$. For $F(s, 2s)$ one has $M = 3s$. □

A by-product is a structural statement about three speeds: a triple (a, b, c) with $\mathrm{ML}(a, b, c) < 1/3$ satisfies, up to order, $c = 2b$ or $c = a + b$ after dividing by the common divisor (and in the first case the triple is $\{1, 2, 3s\}$ by Lemma 9.2), since otherwise $T_{(a,b,c)} \times \mathbb{T}$ would be a subtorus with $\mathrm{ML} < 1/3$ lying in no special three-dimensional subtorus, and Proposition 9.5 together with Lemma 9.1 excludes this.

Corollary 9.7. *The set of accumulation points of the set of values* $\mathrm{ML}(V)$*, over all quadruples* V *of distinct positive integers, that lie in* $(0, \frac{1}{3})$ *is* $\{s/(3s + 1) : s \geq 1\}$*, the three-runner spectrum below* $\frac{1}{3}$*.*

Proof. Giri and Kravitz prove [12, Theorem 1.4] that every accumulation point of the four-runner spectrum is the value of a subtorus of

dimension at least two, and [12, Theorem 1.3] (from [15]) that every three-runner value is an accumulation point. Two remarks reconcile the conventions. First, [12] allows repeated speeds. A tuple with a repeated speed has the value of a set of at most three distinct speeds, and those values in $(0, \frac{1}{3})$ form the discrete set $\{s/(3s+1)\}$ by [15, Theorems 5.1, 6.1 and 7.2], so a sequence of such values converging to a point of $(0, \frac{1}{3})$ is eventually constant, and the accumulation points in $(0, \frac{1}{3})$ are the same whether or not repeated speeds are allowed. Second, subtori contained in a coordinate hyperplane have value 0 and three-dimensional subtori have value at least $\frac{1}{3}$ by Lemma 4.1, so in $(0, \frac{1}{3})$ the accumulation points lie in the set of values of coordinate-free two-dimensional subtori, which by Theorem 9.6 is the three-runner spectrum below $\frac{1}{3}$. □

In the notation of [12], with $D = \frac{1}{2} - \mathrm{ML}$, this says $\mathrm{acc}(\mathcal{S}(4)) \cap (\frac{1}{6}, \frac{1}{2}) = \mathcal{S}(3) \cap (\frac{1}{6}, \frac{1}{2})$, which is the case $n = 4$ of [12, Conjecture 1.5] on the interval $(\frac{1}{6}, \frac{1}{2})$. The next subsection extends this to $D \geq \frac{1}{10}$.

9.1. **From one third to two fifths.** The method reaches the next accumulation point $\frac{2}{5} = \mathrm{ML}(1,4) = \mathrm{ML}(2,3)$ of the three-runner spectrum. Two ingredients are new. Lemma 9.9 replaces the uniform bound $A^{-1/2}$ of Lemma 3.1 by a bound in terms of a shortest character of the subtorus, which is much stronger when that character has a large ℓ^1 norm. Lemma 9.13 determines the values of all two-dimensional subtori of the five three-dimensional subtori with loneliness $\frac{2}{5}$, which carry the infinitely many values that accumulate at $\frac{2}{5}$. Put

$$\begin{aligned}\mathcal{R} &= \Big\{ \frac{p}{q} \in \Big[\frac{1}{3}, \frac{2}{5}\Big) : \ \gcd(p,q) = 1, \ 1 \leq 2q - 5p \leq 4 \Big\} \\ &= \Big\{ \frac{1}{3}, \frac{6}{17}, \frac{5}{14}, \frac{4}{11}, \frac{7}{19}, \frac{10}{27}, \frac{3}{8}, \frac{14}{37}, \dots \Big\}.\end{aligned}$$

For $p/q \in \mathcal{R}$ with $r = 2q - 5p$ we have $q \equiv 3r \pmod 5$, so $(p,q) = (2j+1, 5j+3)$, $(2j, 5j+1)$, $(2j+1, 5j+4)$ or $(2j, 5j+2)$ for $r = 1, 2, 3, 4$. Since $\frac{2j+1}{5j+3} = \frac{2J}{5J+2}$ with $J = 4j+2$ and $\frac{2j}{5j+1} = \frac{2J}{5J+2}$ with $J = 2j$, and since conversely $\frac{2j}{5j+2}$ and $\frac{2j+1}{5j+4}$ lie in $[\frac{1}{3}, \frac{2}{5})$ for $j \geq 2$ and have $2q - 5p$ dividing 4 and 3 in lowest terms,

$$\mathcal{R} = \Big\{ \frac{2j}{5j+2} : j \geq 2 \Big\} \cup \Big\{ \frac{2j+1}{5j+4} : j \geq 2 \Big\}. \tag{5}$$

Theorem 9.8. (i) *Every coordinate-free two-dimensional subtorus U of $\mathbb{T}^4$ with $\mathrm{ML}(U) < \frac{2}{5}$ has $\mathrm{ML}(U) \in \{\frac{s}{3s+1} : s \geq 1\} \cup \mathcal{R}$.*

(ii) *The values of* ML *on triples of positive integers below* $\frac{2}{5}$ *are exactly the numbers* $\frac{s}{3s+1}$, $s \geq 1$, *and the elements of* $\mathcal{R}$. *For* $j \geq 2$,

$$\mathrm{ML}(2,3,5j) = \frac{2j}{5j+2}, \qquad \mathrm{ML}(1,4,5j+3) = \frac{2j+1}{5j+4}.$$

Lemma 9.9 (Shortest character)**.** *Let* U *be a two-dimensional subtorus of* $\mathbb{T}^4$ *with* $A = \mathrm{covol}(U)$, *and let* $L = \{m \in \mathbb{Z}^4 : m \cdot \lambda = 0 \text{ for all } \lambda \in \Lambda_U\}$ *be its lattice of characters. Then* L *is a primitive lattice of rank two with* $\mathrm{covol}(L) = A$, *and for every basis* m_1, m_2 *of* L *we have* $U \subseteq W_{m_1}$ *and*

$$\mathrm{ML}(U) \;\geq\; \mathrm{ML}(W_{m_1}) - \frac{|m_1|}{2A}.$$

Proof. $L = \mathbb{Z}^4 \cap E^\perp$ with $E = \mathbb{R}\Lambda_U$, so L is primitive of rank two. Every $x \in \mathbb{Z}^4$ satisfies $\pi(x) \cdot m = x \cdot m \in \mathbb{Z}$ for $m \in L$, and since L is primitive every homomorphism $L \to \mathbb{Z}$ is of the form $m \mapsto x \cdot m$ with $x \in \mathbb{Z}^4$. Hence $\pi(\mathbb{Z}^4)$ is the dual lattice of L in $E^\perp$, and Lemma 3.1(a) gives $\mathrm{covol}(L) = 1/\mathrm{covol}(\pi(\mathbb{Z}^4)) = A$. Choose $w \in \mathbb{Z}^4$ with $m_1 \cdot w = 0$ and $m_2 \cdot w = 1$. Then $w \notin \Lambda_U$, and the subtorus S' with $\mathbb{R}\Lambda_{S'} = E + \mathbb{R}w$ has character lattice $\{m \in L : m \cdot w = 0\} = \mathbb{Z}m_1$, so $S' = W_{m_1}$. The vector $\pi(w) \in E^\perp$ satisfies $\pi(w) \cdot m_1 = 0$ and $\pi(w) \cdot m_2 = 1$, so $|\pi(w)| = 1/h$, where h is the distance from m_2 to $\mathbb{R}m_1$. As $A = |m_1|h$, we get $|\pi(w)|_\infty \leq |\pi(w)| = |m_1|/A$, and Lemma 3.1(c) gives the inequality. □

Lemma 9.10. *Let* $m \in \mathbb{Z}^4$ *be primitive with* $|m|_1 = \sum_i |m_i| \geq 2$. *Then* $\mathrm{ML}(W_m) = \frac{1}{2}$ *if* $|m|_1$ *is even, and* $\mathrm{ML}(W_m) = \frac{s}{2s+1}$ *if* $|m|_1 = 2s+1$. *If* $|m|_1 = 5$, *then, up to symmetry,* W_m *is one of*

$$W_A = \{(y, 4y, z, u)\}, \quad W_B = \{(2y, 3y, z, u)\},$$
$$W_C = \{(a, b, 3a+b, u)\},$$
$$W_D = \{(a, b, 2a+2b, u)\}, \quad W_E = \{(a, b, c, 2a+b+c)\},$$

where the parameters run over $\mathbb{T}$.

Proof. Put $x_i = \mathrm{sgn}(m_i)\, t$ for $m_i \neq 0$ and $x_i = \frac{1}{2}$ for $m_i = 0$, with $t = \lfloor \frac{|m|_1}{2} \rfloor / |m|_1$. Then $m \cdot x = \lfloor \frac{|m|_1}{2} \rfloor \in \mathbb{Z}$ and $G(x) = t$, which is $\frac{1}{2}$ or $\frac{s}{2s+1}$. Conversely let $|m|_1 = 2s+1$, $c = \frac{s}{2s+1}$, and suppose $\|x_i\| > c$ for all i with $m_i \neq 0$. Then $y_i = \mathrm{sgn}(m_i)x_i$ has a representative in $(c, 1-c)$, and $m \cdot x = \sum_i |m_i| y_i$ lies in $((2s+1)c, (2s+1)(1-c)) = (s, s+1)$, so $x \notin W_m$. Hence $\mathrm{ML}(W_m) \leq c$. If $|m|_1 = 5$, then up to signed permutations the absolute values of the entries of m are $(4,1,0,0)$, $(3,2,0,0)$, $(3,1,1,0)$, $(2,2,1,0)$ or $(2,1,1,1)$, and the five parametrizations are injective homomorphisms $\mathbb{T}^3 \to \mathbb{T}^4$ onto W_m

for $m = (4,-1,0,0)$, $(3,-2,0,0)$, $(3,1,-1,0)$, $(2,2,-1,0)$, $(2,1,1,-1)$. □

Lemma 9.11. *Let U be a coordinate-free two-dimensional subtorus with $\mathrm{ML}(U) < \frac{2}{5}$ that is contained in no W_m with $|m|_1 \in \{3,5\}$. Then $\mathrm{covol}(U) < \frac{35}{2}\sqrt{37}$, that is, $\mathrm{covol}(U)^2 \le 11{,}331$.*

Proof. Let $A = \mathrm{covol}(U)$ and let m_1 be a shortest nonzero vector of L. It is part of a basis of L and primitive in $\mathbb{Z}^4$. As U is coordinate-free, $m_1 \neq \pm e_i$, so $|m_1|_1 \ge 2$, and $|m_1|_1 \notin \{3,5\}$ by hypothesis. Hermite's bound in rank two gives $|m_1|^2 \le \frac{2}{\sqrt{3}}A$, and Lemmas 9.9 and 9.10 give $\frac{2}{5} > \mathrm{ML}(U) \ge \mathrm{ML}(W_{m_1}) - |m_1|/(2A)$. If $|m_1|_1$ is even, this gives $A < 5|m_1|$, hence $|m_1| < 10/\sqrt{3}$ and $A < 50/\sqrt{3}$. If $|m_1|_1 = 7$, it gives $A < \frac{35}{2}|m_1| \le \frac{35}{2}\sqrt{37}$, since $|m|^2 \le 37$ for primitive m with $|m|_1 = 7$. If $|m_1|_1 = 2s+1 \ge 9$, it gives $A < \frac{5(2s+1)}{2(s-2)}|m_1|$. For $s \le 6$ use $|m_1| \le |m_1|_1$ to get $A < \frac{5(2s+1)^2}{2(s-2)} \le \frac{845}{8}$. For $s \ge 7$, Hermite's bound gives $|m_1| < \frac{5(2s+1)}{\sqrt{3}(s-2)}$, while $|m_1| \ge \frac{1}{2}|m_1|_1 = \frac{2s+1}{2}$, so $s = 7$ and $A < \frac{15}{2}\cdot\frac{15}{\sqrt{3}} < 65$. In every case $A < \frac{35}{2}\sqrt{37}$, and $(\frac{35}{2})^2 \cdot 37 = 11{,}331.25$. □

We now treat the five subtori of Lemma 9.10. For $\tau \in \{A,B,C,D,E\}$ let $\varphi_\tau : \mathbb{T}^3 \to W_\tau$ be the parametrization of Lemma 9.10 and $G_\tau = G \circ \varphi_\tau$. As in Section 5, the two-dimensional subtori of W_τ are the images of $U_n = \{y \in \mathbb{T}^3 : n \cdot y \in \mathbb{Z}\}$ for primitive $n \in \mathbb{Z}^3$. Write $J(\varepsilon) = [\frac{2}{5} - \varepsilon, \frac{3}{5} + \varepsilon]$ and, for $0 \le \varepsilon \le \frac{1}{15}$,

$$P_A(\varepsilon) = [\tfrac{2}{5} - \varepsilon, \tfrac{2}{5} + \tfrac{\varepsilon}{4}] \times J(\varepsilon)^2, \quad P_B(\varepsilon) = [\tfrac{1}{5} - \tfrac{\varepsilon}{2}, \tfrac{1}{5} + \tfrac{\varepsilon}{3}] \times J(\varepsilon)^2,$$
$$P_C(\varepsilon) = \big((\tfrac{2}{5}, \tfrac{2}{5}) + \varepsilon\Delta_C\big) \times J(\varepsilon), \quad P_D(\varepsilon) = \big((\tfrac{2}{5}, \tfrac{2}{5}) + \varepsilon\Delta_D\big) \times J(\varepsilon),$$
$$P_E(\varepsilon) = (\tfrac{2}{5}, \tfrac{2}{5}, \tfrac{2}{5}) + \varepsilon\Delta_E,$$

where

$$\Delta_C = \mathrm{conv}\{(-1,-1), (\tfrac{2}{3}, -1), (-1, 4)\},$$
$$\Delta_D = \mathrm{conv}\{(-1,-1), (\tfrac{3}{2}, -1), (-1, \tfrac{3}{2})\},$$
$$\Delta_E = \mathrm{conv}\{(-1,-1,-1), (\tfrac{3}{2}, -1, -1), (-1, 4, -1), (-1, -1, 4)\}.$$

Lemma 9.12. *For $0 \le \varepsilon \le \frac{1}{15}$ the set $\{y \in \mathbb{T}^3 : G_\tau(y) \ge \frac{2}{5} - \varepsilon\}$ is the image of $P_\tau(\varepsilon) \cup (-P_\tau(\varepsilon))$.*

Proof. Put $d = \frac{2}{5} - \varepsilon \in [\frac{1}{3}, \frac{2}{5}]$. The conditions $\|z\|, \|u\| \ge d$ mean $z, u \in [d, 1-d] + \mathbb{Z}$, which is $J(\varepsilon) + \mathbb{Z}$, and $J(\varepsilon) = -J(\varepsilon) + 1$. For W_A: if $y \in [0, \frac{1}{2}]$, then $\|y\| \ge d$ means $y \ge d$, and then $4y \in [\frac{4}{3}, 2]$, so $\|4y\| \ge d$ means $4y \in [1+d, 2-d]$. As $d \ge \frac{1}{3}$, $\frac{1+d}{4} \le d$, so the solutions in $[0, \frac{1}{2}]$ form $[d, \frac{2-d}{4}] = [\frac{2}{5} - \varepsilon, \frac{2}{5} + \frac{\varepsilon}{4}]$, and those in $[\frac{1}{2}, 1]$ are

their negatives. For W_B: in $[0,\frac12]$, $\|2y\|\ge d$ means $y\in[\frac d2,\frac{1-d}{2}]$, and $\|3y\|\ge d$ means $y\in[\frac d3,\frac{1-d}{3}]\cup[\frac{1+d}{3},\frac12]$. Since $\frac{1+d}{3}>\frac{1-d}{2}$ for $d>\frac15$, the intersection is $[\frac d2,\frac{1-d}{3}]=[\frac15-\frac{\varepsilon}{2},\frac15+\frac{\varepsilon}{3}]$. For W_C, W_D, W_E let ℓ be the last coordinate form, $3a+b$, $2a+2b$ or $2a+b+c$. Its coefficients are at least 1 and sum to 4. On the box where the other coordinates lie in $[d,1-d]$, ℓ takes values in $[4d,4-4d]\subseteq[\frac43,\frac83]$, so $\|\ell\|\ge d$ means $\ell\in[1+d,2-d]\cup[2+d,3-d]$, and the bounds $1+d\le 4d$ and $4-4d\le 3-d$ are automatic. The set $\{\text{coordinates}\ge d,\ \ell\le 2-d\}$ already forces every coordinate to be at most $2-d-3d\le 1-d$, and the substitution $y\mapsto -y$ maps it onto $\{\ell\ge 2+d\}$ modulo $\mathbb{Z}^3$. Writing the coordinates as $\frac25+\varepsilon\delta_i$, the conditions read $\delta_i\ge -1$ and $3\delta_1+\delta_2\le 1$, $\delta_1+\delta_2\le\frac12$, respectively $2\delta_1+\delta_2+\delta_3\le 1$, which define Δ_C, Δ_D, Δ_E. $\square$

Lemma 9.13 (Subtori of the five tori)**.** *Let $\tau\in\{A,B,C,D,E\}$ and let $n\in\mathbb{Z}^3$ be primitive with $\varphi_\tau(U_n)$ coordinate-free. If $\frac13\le \mathrm{ML}(U_n)<\frac25$, then $\mathrm{ML}(U_n)\in\mathcal{R}$.*

Proof. Write $V=\mathrm{ML}(U_n)=\frac25-\varepsilon^*$. Since $P_\tau(\varepsilon)$ is convex and U_n lifts to $\{y\in\mathbb{R}^3 : n\cdot y\in\mathbb{Z}\}$, Lemma 9.12 shows that for $\varepsilon\le\frac{1}{15}$ we have $V\ge\frac25-\varepsilon$ exactly when the interval $n\cdot P_\tau(\varepsilon)$ contains an integer (the set $-P_\tau$ gives the negative interval). Each $P_\tau(\varepsilon)$ is a product of intervals $[c-\varepsilon e,c'+\varepsilon e']$ and of a set $p+\varepsilon\Delta$, so

$$n\cdot P_\tau(\varepsilon)=[\alpha-\varepsilon a,\ \beta+\varepsilon b],$$

where $\beta=n\cdot x_0$ and $b=n\cdot w$ for a vertex $x_0+\varepsilon w$ of $P_\tau(\varepsilon)$ at which $n\cdot y$ is maximal for every $\varepsilon\in(0,\frac{1}{15}]$, and similarly α, a at a minimal vertex. Here $x_0\in\frac15\mathbb{Z}^3$, so $\alpha,\beta\in\frac15\mathbb{Z}$. As $V<\frac25$, $[\alpha,\beta]$ contains no integer. Let $N=\lceil\beta\rceil$, $N'=\lfloor\alpha\rfloor$, $k=5(N-\beta)$ and $k'=5(\alpha-N')$, so $k,k'\in\{1,2,3,4\}$, and $\varepsilon^*=\min(\frac{k}{5b},\frac{k'}{5a})$ with $\frac{k}{0}=\infty$. Suppose first that the minimum is $\frac{k}{5b}$.

At the vertex $x_0+\varepsilon w$ three linearly independent coordinate forms ϕ_i, $i\in I$, of φ_τ are active, that is, $\sigma_i\phi_i(x_0+\varepsilon w)=\kappa_i+\frac25-\varepsilon$ with $\sigma_i=\pm1$ and $\kappa_i\in\mathbb{Z}$. Write $n=\sum_{i\in I}\lambda_i\sigma_i\phi_i$, so that $\lambda_i\in\frac1D\mathbb{Z}$ with $D=|\det(\phi_i)_{i\in I}|$, and put $K=\sum\lambda_i\kappa_i$ and $\Lambda=\sum\lambda_i$. Evaluating at $\varepsilon=0$ and $\varepsilon=\varepsilon^*$ gives $\beta=K+\frac25\Lambda$ and $N=K+V\Lambda$. Hence $V=P/Q$ with $P=D(N-K)$ and $Q=D\Lambda$ integers and $2Q-5P=5D(\beta-N)=-Dk$. As $V<\frac25$, writing $V=p/q$ in lowest terms gives $1\le 2q-5p\le Dk$. If the minimum is $\frac{k'}{5a}$, the same argument for $-n$, which exchanges the two ends, gives $1\le 2q-5p\le D'k'$, where D' is the determinant at the minimal vertex. So $V\in\mathcal{R}$ whenever the end attaining the minimum has determinant 1.

The vertices with $D > 1$ are the following. For $\tau = A$ the ends $y = \frac{2}{5} + \frac{\varepsilon}{4}$ (form $4y$, $D = 4$). For $\tau = B$ the ends $y = \frac{1}{5} - \frac{\varepsilon}{2}$ ($2y$, $D = 2$) and $y = \frac{1}{5} + \frac{\varepsilon}{3}$ ($3y$, $D = 3$). For $\tau = C$ the vertex $(\frac{2}{3}, -1)$ ($D = 3$). For $\tau = D$ the vertices $(\frac{3}{2}, -1)$, $(-1, \frac{3}{2})$ ($D = 2$). For $\tau = E$ the vertex $(\frac{3}{2}, -1, -1)$ ($D = 2$). We show that if an end at such a vertex attains the minimum and the product of its determinant and its k exceeds 4, then the other end attains the minimum as well, at a vertex with determinant 1, or with determinant 2 and $k \le 2$. Since V does not depend on which end is used, this proves the lemma.

Types A and B. Using the symmetries $y \mapsto -y$, $z \mapsto -z$, $u \mapsto -u$ we may take $n_1, n_2, n_3 \ge 0$, and $c = n_2 + n_3 \ge 1$ since $\varphi_\tau(U_n)$ is coordinate-free. If $n_1 = 0$, then $\varphi_\tau(U_n)$ is the product of the circle of $(1, 4)$ or $(2, 3)$ with a circle in the (z, u) plane, so $V = \min(\frac{2}{5}, \mathrm{ML}(n_2, n_3))$, and $V \in [\frac{1}{3}, \frac{2}{5})$ forces $V = \frac{1}{3} \in \mathcal{R}$. Let $n_1 \ge 1$. The interval $[\alpha, \beta]$ has length $\frac{c}{5}$, so $k + k' = 5 - c$. For A, $a = n_1 + c$, $b = \frac{n_1}{4} + c$, and the upper end has $D = 4$, while the lower end has $D = 1$. If $k \ge 2$, then $k' \le 3 - c$, and $ka \ge 2n_1 + 2c > (3 - c)(\frac{n_1}{4} + c) \ge k'b$, so $\frac{k}{5b} > \frac{k'}{5a}$ and the upper end is not the minimum. For B, $a = \frac{n_1}{2} + c$ and $b = \frac{n_1}{3} + c$. If the upper end ($D = 3$) gives the minimum with $k \ge 2$, then $k' \le 3 - c$ and $ka \ge n_1 + 2c > (3 - c)(\frac{n_1}{3} + c) \ge k'b$, a contradiction. If the lower end ($D = 2$) gives the minimum with $k' \ge 3$, then $c = 1$, $k = 1$, $k' = 3$, and $k'b = n_1 + 3 > \frac{n_1}{2} + 1 = ka$, again a contradiction.

Types C, D, E. Let f be the linear form n restricted to the δ-coordinates of Δ_τ, and $M = \max_{\Delta_\tau} f$, $m = -\min_{\Delta_\tau} f$. As 0 is an interior point of Δ_τ and $f \ne 0$ (otherwise $\varphi_\tau(U_n)$ lies in $\{u = 0\}$), $M, m > 0$. With $n_3 \ge 0$ for C and D, we have $b = M + c$, $a = m + c$ with $c = n_3$ for C, D and $c = 0$ for E, and $k + k' = 5 - c$. The origin is the barycenter of the vertices of Δ_τ with weights $(\frac{1}{5}, \frac{3}{5}, \frac{1}{5})$, $(\frac{1}{5}, \frac{2}{5}, \frac{2}{5})$ and $(\frac{1}{5}, \frac{2}{5}, \frac{1}{5}, \frac{1}{5})$ in the order listed, so the weighted sum of the values of f at the vertices vanishes. If f is maximal at the vertex with $D > 1$ and weight μ, this gives $\mu M \le (1 - \mu)m$, that is, $M \le \frac{2}{3}m$ for C and $M \le \frac{3}{2}m$ for D and E. For C ($D = 3$), suppose $k \ge 2$ and $\frac{k}{b} \le \frac{k'}{a}$. Since $b \le a$, this needs $k \le k'$, so $k + k' \ge 4$ and $c \le 1$. If $c = 1$ then $k = k' = 2$ and $a \le b$ forces $m \le M \le \frac{2}{3}m$, which is impossible. If $c = 0$ then $(k, k') = (2, 3)$ and $2m \le 3M \le 2m$, so the two ends give the same ε^*, and the lower end is a vertex with $D = 1$. For D and E ($D = 2$), suppose $k \ge 3$ and $ka \le k'b$. Then $k' = 5 - c - k \le 2 - c$. If $c = 1$, then $k = 3$, $k' = 1$ and $3(m + 1) \le M + 1 \le \frac{3}{2}m + 1$, which is impossible. So $c = 0$, $k' = 5 - k \le 2$, and $3m \le km \le k'M \le \frac{3}{2}k'm \le 3m$. Hence $k = 3$, $k' = 2$ and the two ends give the same ε^*. The lower end is a vertex with $D = 1$, or for $\tau = D$ possibly $(-1, \frac{3}{2})$, with $D = 2$ and

$Dk' = 4$. The case of a minimal vertex with $D > 1$ follows by replacing n with $-n$. □

Proposition 9.14. *There are* 789,460,684 *coordinate-free primitive rank-two sublattices of* $\mathbb{Z}^4$ *with squared covolume at most* 11,331*. Of these,* 21,301,056 *are contained in a signed permutation image of* W^1 *or* W^2*, and* 36,832,896 *further ones in some* W_m *with* $|m|_1 = 5$*. Each of the remaining ones except* 18,960 *has a point with* $G \geq \frac{2}{5}$*: a grid point* $(i/Q, j/Q)$ *with* $Q \in \{2, 10, 20, 30\}$*, or else the point returned by the exact maximization of Section 12, at which* G *is evaluated exactly. The remaining* 18,960 *subtori have* $\mathrm{ML} < \frac{2}{5}$*. They have squared covolume at most* 1,250*, form* 141 *orbits under signed coordinate permutations, and their values are* $\frac{7}{19}$*,* $\frac{3}{8}$*,* $\frac{8}{21}$*,* $\frac{5}{13}$*,* $\frac{12}{31}$*,* $\frac{9}{23}$*,* $\frac{13}{33}$ *and* $\frac{15}{38}$*, all of them in* $\mathcal{R}$*.*

Proof of Theorem 9.8. (i) Let $\mathrm{ML}(U) < \frac{2}{5}$. If $\mathrm{ML}(U) < \frac{1}{3}$, Theorem 9.6 gives $\mathrm{ML}(U) = \frac{s}{3s+1}$. Let $\frac{1}{3} \leq \mathrm{ML}(U) < \frac{2}{5}$. If U lies in a signed permutation image of W^1 or W^2, then $\mathrm{ML}(U) \leq \frac{1}{3}$, so $\mathrm{ML}(U) = \frac{1}{3} \in \mathcal{R}$. If U lies in some W_m with $|m|_1 = 5$, then up to symmetry $U = \varphi_\tau(U_n)$ by Lemma 9.10, and Lemma 9.13 applies. Otherwise $\mathrm{covol}(U)^2 \leq 11{,}331$ by Lemma 9.11, and Proposition 9.14 applies.

(ii) For a triple v of distinct positive integers, $T_v \times \mathbb{T}$ is a coordinate-free two-dimensional subtorus with $\mathrm{ML}(T_v \times \mathbb{T}) = \mathrm{ML}(v)$, so by (i) every value below $\frac{2}{5}$ is $\frac{s}{3s+1}$ or lies in $\mathcal{R}$. A triple with a repeated entry has the value of a pair, $\frac{s}{2s+1}$ or $\frac{1}{2}$, and only $\frac{1}{3} \in \mathcal{R}$ occurs below $\frac{2}{5}$. Conversely, $\frac{s}{3s+1} = \mathrm{ML}(1, 2, 3s)$ by [15, Theorem 5.1], and by (5) it remains to prove the two formulas. The product $T_{(1,4,k)} \times \mathbb{T}$ is $\varphi_A(U_{(k,1,0)})$ up to symmetry, and $(k, 1, 0) \cdot P_A(\varepsilon) = [\frac{2k+2}{5} - \varepsilon(k+1),\ \frac{2k+3}{5} + \varepsilon(\frac{k}{4}+1)]$. For $k = 5j+3$ this is $[2j + \frac{8}{5} - \varepsilon(k+1),\ 2j + \frac{9}{5} + \varepsilon(\frac{k}{4}+1)]$, which first contains an integer at $\varepsilon = \min(\frac{4}{5(k+4)}, \frac{3}{5(k+1)}) = \frac{3}{5(k+1)}$ for $k \geq 8$, a value at most $\frac{1}{15}$. By Lemma 9.12, $\mathrm{ML}(1, 4, 5j+3) = \frac{2}{5} - \frac{3}{5(5j+4)} = \frac{2j+1}{5j+4}$. Similarly $T_{(2,3,k)} \times \mathbb{T} = \varphi_B(U_{(k,1,0)})$, and for $k = 5j$ the interval $(k, 1, 0) \cdot P_B(\varepsilon) = [j + \frac{2}{5} - \varepsilon(\frac{k}{2}+1),\ j + \frac{3}{5} + \varepsilon(\frac{k}{3}+1)]$ first contains an integer at $\varepsilon = \frac{4}{5(k+2)} \leq \frac{1}{15}$ for $j \geq 2$, so $\mathrm{ML}(2, 3, 5j) = \frac{2}{5} - \frac{4}{5(5j+2)} = \frac{2j}{5j+2}$. □

Corollary 9.15. (i) *The accumulation points of the four-runner spectrum in* $(0, \frac{2}{5}]$ *are exactly the three-runner values there,* $\{\frac{s}{3s+1} : s \geq 1\} \cup \mathcal{R} \cup \{\frac{2}{5}\}$*. In the notation of* [12]*,*

$$\mathrm{acc}(\mathcal{S}(4)) \cap [\tfrac{1}{10}, \tfrac{1}{2}) = \mathcal{S}(3) \cap [\tfrac{1}{10}, \tfrac{1}{2}),$$

the case $n = 4$ *of* [12, Conjecture 1.5] *on* $[\frac{1}{10}, \frac{1}{2})$*.*

(ii) $\frac{7}{50} \notin \mathcal{S}_2(4)$*. Since* $\frac{7}{50} \in \mathcal{S}^*(3) = \mathcal{S}_2^*(4)$ *by* [12, Proposition 8.1 and Lemma 3.2]*, the spectrum of two-dimensional subtori of* $\mathbb{T}^4$

is strictly smaller than the spectrum of two-dimensional closed subgroups.

(iii) *In the notation of* [14], *with* $\mathrm{Prog}(\alpha,\beta)=\{\alpha s+\beta : s\ge 0\}$ *and* $\frac{c}{5}\,\mathrm{Prog}(\alpha,\beta)$ *read as* $\{\frac{c}{5x} : x\in \mathrm{Prog}(\alpha,\beta)\}$,

$$\mathcal{S}_1(3)\cap(\tfrac{1}{10},\tfrac{1}{6}]=\Big(\big(\tfrac{1}{10}+\tfrac{4}{5}\,\mathrm{Prog}(5,7)\big)\cup\big(\tfrac{1}{10}+\tfrac{3}{5}\,\mathrm{Prog}(5,9)\big)\Big)\cap(\tfrac{1}{10},\tfrac{1}{6}].$$

So the finite symmetric difference in [14, Theorem 1.4] *has no element in* $(\frac{1}{10},\frac{1}{6}]$. *It consists of the single value* $\frac{3}{14}$, *the term* $s=0$ *of the first progression, which lies outside that interval.*

Proof. (i) Let A be the set of accumulation points in $(0,\frac{2}{5}]$ of the values of all quadruples, repeated speeds allowed, and A' the same for quadruples of distinct integers. Then $A'\subseteq A$. Every point of A is the value of a two-dimensional subtorus by [12, Theorem 1.4], which is coordinate-free since the value is positive, so it is $\frac{2}{5}$ or lies in the set of Theorem 9.8(i). Every three-runner value, in particular $\frac{2}{5}=\mathrm{ML}(1,4,6)$, is a lower accumulation point of values of quadruples of distinct integers by [12, Theorem 1.3], whose proof, from [15], adds one large fourth speed to a triple. With Theorem 9.8(ii) this gives $A=A'=\mathcal{S}(3)\cap(0,\frac{2}{5}]$. (ii) $\frac{9}{25}\notin\mathcal{R}$ since $2\cdot 25-5\cdot 9=5$, and a two-dimensional subtorus contained in a coordinate hyperplane has value 0. (iii) Since $D=\frac{1}{2}-\mathrm{ML}$, we have $\frac{1}{2}-\frac{2j}{5j+2}=\frac{1}{10}+\frac{4}{5(5j+2)}$ and $\frac{1}{2}-\frac{2j+1}{5j+4}=\frac{1}{10}+\frac{3}{5(5j+4)}$, and $5j+2$, $5j+4$ run through $\mathrm{Prog}(5,7)$, $\mathrm{Prog}(5,9)$ for $j\ge 1$. The terms with $j=1$ are $\frac{3}{14}$ and $\frac{1}{6}$. Now apply Theorem 9.8(ii) and (5). □

The same two ingredients apply one level higher. By Lemmas 9.9 and 9.10, a coordinate-free two-dimensional subtorus with $\mathrm{ML}<\frac{3}{7}$ lies in some W_m with $|m|_1\in\{3,5,7\}$ or has bounded covolume, and the ten types of W_m with $|m|_1=7$ all have loneliness $\frac{3}{7}$. The contact argument of Lemma 9.13 gives $1\le 3q-7p\le 6$ at vertices with determinant 1, but it no longer excludes the other vertices: the subtorus $U=\{(10X+Y,\,-49X-5Y,\,9X,\,Y)\}$ of $W_{(4,1,1,1)}$ has $\mathrm{ML}(U)=\frac{43}{103}$, and $3\cdot 103-7\cdot 43=8$. This value is again a three-runner value, $\frac{43}{103}=\mathrm{ML}(1,102,25)$, so the level $\frac{3}{7}$ needs a finer identification of the three-runner values than (5). We have not carried this out.

Remark 9.16 (Why slicing at a maximizer does not suffice). Giri and Kravitz observe that proving $\mathcal{S}_2(4)=\mathcal{S}(3)$ would settle [12, Conjecture 1.5] for $n=4$, and that the slicing argument behind $\mathcal{S}_2(n)\subseteq\mathcal{S}^*(n-1)$ would have to be refined. Let x^* be a maximizer of G on U and let a,b be active coordinates, $\|x_a^*\|=\|x_b^*\|=\mathrm{ML}(U)$, with $x_a^*=\sigma x_b^*$. The subgroup $U\cap\{x_a=\sigma x_b\}$ contains x^*, and if x^* lies

on its identity component, a circle, then $\mathrm{ML}(U)$ is the loneliness of a circle with a repeated speed, a three-runner value. For all 916,492 coordinate-free two-dimensional subtori of covolume at most 20, every one of the 1,348,608 maximizers at which two independent coordinate forms are active, over the 582,624 subtori with $\frac{1}{3} < \mathrm{ML}(U) < \frac{1}{2}$, lies on such a circle. This fails in general. The subtorus

$$U = \{(y_1, y_2, y_3, u) : 4y_1 + 21y_2 + 5y_3 = 0,\ u = 2y_1 + 3y_2 + 3y_3\}$$

has $\mathrm{ML}(U) = \frac{13}{30}$, its only maximizers are $\pm x^*$ with $x^* = (\frac{13}{30}, \frac{13}{30}, \frac{13}{30}, \frac{7}{15})$, and x^* lies on a non-identity component of each of its three slices. Indeed, with $\lambda = (4, 21, 5)$ and c the index other than a, b, the slice $U \cap \{y_a = y_b\}$ has $\gcd(\lambda_c, 30) \in \{2, 3, 5\}$ components, the point $(\frac{r}{30}, \frac{r}{30}, \frac{r}{30}, \frac{8r}{30})$ of U lies on the component indexed by r modulo $\gcd(\lambda_c, 30)$, and x^* is the point with $r = 13$. Nevertheless $\frac{13}{30} = \mathrm{ML}(1, 8, 29)$, and the circle of $(1, 29, 8)$ at the times $r/30$ reproduces the values of G at these thirty points, which form the finite subgroup $U \cap \{y_1 = y_2 = y_3\}$. Among the 19,000 distinct subtori $\{(y, u) : \lambda \cdot y = 0,\ \mu u = \theta \cdot y\}$ with $\lambda \in \mathbb{Z}^3_{>0}$ primitive, $\lambda_1 + \lambda_2 + \lambda_3 \le 42$, $\gcd(\lambda_c, \lambda_1 + \lambda_2 + \lambda_3) > 1$ for every c, $\theta \in \{-4, \dots, 4\}^3$ and $1 \le \mu \le 3$, the 602 with no maximizer on a slice circle take 14 distinct values, all of them three-runner values. So a proof of $\mathcal{S}_2(4) = \mathcal{S}(3)$ needs a second mechanism besides slicing, such as the circle through this finite subgroup. Below $\frac{2}{5}$ such a mechanism is the arithmetic of the contact vertex in Lemma 9.13, which identifies the values without producing a circle. Section 10 shows that the failure is not isolated: it occurs in an infinite family with values tending to $\frac{1}{2}$, with a loss of at least N^{-3} for every circle with a repeated speed.

10. Slicing obstructions of unbounded order

Remark 9.16 exhibits one subtorus whose maximizers lie off every slice circle. This section gives an infinite family with the same property, in which the slices through the maximizers have arbitrarily many components and the values tend to $\frac{1}{2}$, and it makes the loss quantitative. Call a coordinate-free circle contained in some $\{x_i = \pm x_j\}$, $i \ne j$, a *repeated-speed circle*. For positive integers A, B, C with $\gcd(A, B, C) = 1$ put $\lambda = (A, B, C)$, $N = A + B + C$, $H = N/2$ when N is even, and

$$U(A, B, C) = \{(y_1, y_2, y_3, u) \in \mathbb{T}^4 :\ Ay_1 + By_2 + Cy_3 = 0,\ u = Hy_2\}.$$

This is the graph of $y \mapsto Hy_2$ over the connected kernel of the primitive character λ on $\mathbb{T}^3$, hence a two-dimensional subtorus, and it is coordinate-free since $A, B, C \ne 0$.

Theorem 10.1. *Let $4 \mid N$ and $N \geq 2(A+C)+2$, and put $r = \frac{N}{2} - 1$ and $L = \frac{r}{N} = \frac{1}{2} - \frac{1}{N}$.*

(i) $\mathrm{ML}(U(A,B,C)) = L$, *and the only maximizers are $\pm p$ with $p = (L, L, L, \frac{1}{2})$. Moreover* $\mathrm{ML}(1, \frac{N}{2}, N-1) = L$.

(ii) *For $\{i,j,k\} = \{1,2,3\}$ the slice $U(A,B,C) \cap \{y_i = y_j\}$ has* $\gcd(\lambda_k, N)$ *components, and p lies on its identity component if and only if* $\gcd(\lambda_k, N) = 1$. *A repeated-speed circle in $U(A,B,C)$ attains L if and only if one of* $\gcd(A,N)$, $\gcd(B,N)$, $\gcd(C,N)$ *equals* 1.

(iii) *If* $\gcd(\lambda_k, N) > 1$ *for $k = 1,2,3$, then* $\mathrm{ML}(T) \leq L - N^{-3}$ *for every repeated-speed circle $T \subseteq U(A,B,C)$.*

Proof. (i) Since $\lambda \cdot (L,L,L) = r$ and $Hr/N = \frac{N}{4} - \frac{1}{2}$, the point p lies on U, and $G(p) = L$. Let $G(y,u) \geq L$ and write $y_i = \frac{1}{2} + z_i$ with $|z_i| \leq \frac{1}{N}$. As $\frac{N}{2} \in \mathbb{Z}$, the number $\lambda \cdot z$ is an integer in $[-1,1]$. If it is ± 1, positivity of λ forces $z_i = \pm\frac{1}{N}$ for all i, that is $(y,u) = \mp p$. If it is 0, then $B|z_2| = |Az_1 + Cz_3| \leq \frac{A+C}{N}$, and since $\frac{H}{2} = \frac{N}{4} \in \mathbb{Z}$ we get $u \equiv Hz_2$ and $\|u\| \leq \frac{A+C}{2B}$. With $S = A + C$ and $B = N - S$, the inequality $\frac{S}{2B} < L$ is equivalent to $N^2 - 2(S+1)N + 2S > 0$, which holds for $N \geq 2S+2$. So this case is impossible. For the triple, $t = \frac{r}{N}$ gives the value L, since $\frac{N}{2} \cdot \frac{r}{N} \equiv \frac{1}{2}$ and $\|(N-1)t\| = \|t\|$. Conversely, if all three distances exceed L at some $t = \frac{1}{2} - \delta$ with $0 \leq \delta < \frac{1}{N}$, then $\|(N-1)t\| = \frac{1}{2} - \|(N-1)\delta\|$ forces $(N-1)\delta < \frac{1}{N}$, and then $\|\frac{N}{2}t\| = \|\frac{N}{2}\delta\| \leq \frac{N}{2}\delta < \frac{1}{2(N-1)} < L$, a contradiction.

(ii) On $y_i = y_j = t$ the relation reads $(N - \lambda_k)t + \lambda_k y_k = 0$, whose kernel in $\mathbb{T}^2$ has $\gcd(N-\lambda_k, \lambda_k) = \gcd(\lambda_k, N) = g$ components, labelled by the value of the primitive character $\frac{N-\lambda_k}{g}t + \frac{\lambda_k}{g}y_k$ in $\frac{1}{g}\mathbb{Z}/\mathbb{Z}$. At p this value is $\frac{r}{g}$, and $\gcd(r,N) = 1$ because $r = \frac{N}{2} - 1$ is odd and prime to $\frac{N}{2}$. So p is on the identity component exactly when $g = 1$. A repeated-speed circle attaining L contains $\pm p$ by (i). At p the last coordinate has distance $\frac{1}{2} \neq L$, and $y_i = -y_j$ would give $\frac{2r}{N} \in \mathbb{Z}$. So such a circle lies in some $\{y_i = y_j\}$ and is the identity component of the slice, which contains p only if $g = 1$. Conversely, if $g = 1$ the slice is a circle through p, coordinate-free because no coordinate of p vanishes.

(iii) None of the twelve equations $x_i = \pm x_j$ holds on all of U, because λ has no zero entry and $H > 1$. So a repeated-speed circle $T \subseteq U$ is the identity component of the one-dimensional group $U \cap \{x_i = \pm x_j\}$ for one of them. For $y_i = \sigma y_j$ the vector w with $w_i = \lambda_k$, $w_j = \sigma\lambda_k$, $w_k = -(\lambda_i + \sigma\lambda_j)$, $w_4 = Hw_2$ lies in Λ_U on that slice. For $y_1 = \sigma u$ and $y_3 = \sigma u$ the vectors $(\sigma HC, C, -(\sigma HA + B), HC)$ and $(-(B +$

$\sigma HC), A, \sigma HA, HA)$ do the same, and $y_2 = \pm u$ forces $y_2 = u = 0$ on the identity component, which is not coordinate-free. So T has speeds of absolute value at most HN, and by Lemma 2.1 its loneliness is a/d with integers a and $1 \le d \le 2HN = N^2$. By (ii), $a/d < L$, so $L - \frac{a}{d} = \frac{rd-aN}{Nd} \ge \frac{1}{N^3}$. □

For instance $A = m$, $C = m+1$, $N = 4m(m+1)(2m+1)K$ and $B = N - 2m - 1$, with $m \ge 2$ and $K \ge 1$, satisfy the hypotheses with $\gcd(\lambda_k, N)$ equal to m, $2m+1$ and $m+1$. So the three slices through each maximizer have arbitrarily many components, none of their identity components meets a maximizer, and the values $L = \frac{1}{2} - \frac{1}{N}$ tend to $\frac{1}{2}$. All of them are three-runner values by (i), as Conjecture 1.5 of [12] predicts. For example $U(2, 55, 3)$ has $L = \frac{29}{60}$ and component orders $2, 5, 3$.

The loss in (iii) is not an artifact of the bound. Inside the same subtori, circles with four distinct speeds come closer to L than any repeated-speed circle can.

Theorem 10.2. *Let $4 \mid N$, $N \ge 12$ and $\ell \ge 1$. Then*

$$\mathrm{ML}\left(1,\ \frac{N}{2},\ 2N\ell-3,\ 3N\ell-\frac{N}{2}-2\right) = \frac{(N-2)\ell - 1}{2N\ell - 2} = L - \frac{1}{N(N\ell-1)}.$$

If moreover $60 \mid N$ and $\ell \ge N+1$, then this circle lies in $U(2, N-5, 3)$ up to signs, and its loneliness exceeds that of every repeated-speed circle in $U(2, N-5, 3)$.

Proof. Put $U_N = U(2, B, 3)$ with $B = N - 5$. The map $(s, t) \mapsto (Bs + 3t,\ s,\ -Bs - 2t,\ Hs)$ is an isomorphism $\mathbb{T}^2 \to U_N$ with inverse $s = y_2$, $t = y_1 + y_3$. For $k \ge 0$ the circle $\{t = ks\}$ has the speeds 1, H, $B + 2k$, $B + 3k$ up to sign, and $k = N\ell - H + 1$ gives the four speeds of the theorem.

Let $0 \le \varepsilon \le \varepsilon_0 = \frac{1}{N(N-1)}$. We claim that the points of U_N with $G \ge L - \varepsilon$ are the points (y, Hy_2) with $y = \pm\big((L, L, L) + \eta\big)$ and η in the triangle $\{2\eta_1 + B\eta_2 + 3\eta_3 = 0,\ \eta_i \ge -\varepsilon\}$. Write $y_i = \frac{1}{2} + z_i$ with $|z_i| \le D = \frac{1}{N} + \varepsilon \le \frac{1}{N-1}$. Then $2z_1 + Bz_2 + 3z_3$ is an integer of absolute value at most $ND < 2$. If it is 0, then $\|u\| \le H|z_2| \le \frac{5HD}{B} \le \frac{5N}{2B(N-1)} < \frac{N-3}{2(N-1)} = L - \varepsilon_0$, where the strict inequality is $N^2 - 13N + 15 > 0$, true for $N \ge 12$. If it is -1, put $z_i = -\frac{1}{N} + \eta_i$. Then $\lambda \cdot \eta = 0$ and $\eta_i \ge -\varepsilon$, the vertices of the triangle are $\varepsilon(H-1, -1, -1)$, $\varepsilon(-1, \frac{5}{B}, -1)$ and $\varepsilon(-1, -1, \frac{N-3}{3})$, so $|\eta_i| \le \frac{1}{N}$ and the upper bounds $z_i \le D$ hold, and $\|u\| = \frac{1}{2} - H|\eta_2| \ge \frac{1}{2} - H\varepsilon \ge L - \varepsilon$. The level $+1$ is the negative. This proves the claim.

At p, where $(s,t) = (\frac{r}{N}, \frac{2r}{N})$, the character $ks - t$ takes the value $\frac{(k-2)r}{N}$, and $(k-2)r \equiv (-H-1)(H-1) = 1 - H^2 \equiv 1 \pmod N$ since $4 \mid N$. On the triangle it changes by $k\eta_2 - \eta_1 - \eta_3$, whose values at the three vertices are $-(k+H-2)\varepsilon$, $\frac{5k+2B}{B}\varepsilon$ and $-(k+\frac{N}{3}-2)\varepsilon$. So the circle meets $\{G \geq L-\varepsilon\}$ exactly when $[\frac{1}{N}-(k+H-2)\varepsilon,\ \frac{1}{N}+\frac{5k+2B}{B}\varepsilon]$ contains an integer, and the negative triangle gives the same condition. The first contact is with 0 at $\varepsilon = \frac{1}{N(k+H-2)} = \frac{1}{N(N\ell-1)} \leq \varepsilon_0$, because reaching 1 needs $\varepsilon \geq \frac{(N-1)B}{N(5k+2B)}$, which is larger. This proves the formula. If $60 \mid N$, then $\gcd(2,N)$, $\gcd(N-5,N)$ and $\gcd(3,N)$ are 2, 5 and 3, and for $\ell \geq N+1$ the deficit $\frac{1}{N(N\ell-1)}$ is smaller than N^{-3}, so Theorem 10.1(iii) gives the comparison. □

So in $U(2,N-5,3)$ with $60 \mid N$ the loneliness of circles with distinct speeds approaches the maximum L, a three-runner value, while every repeated-speed circle stays at distance at least N^{-3} below it. A proof of $\mathcal{S}_2(4) = \mathcal{S}(3)$ by passing from a subtorus to a circle with a repeated speed inside it is therefore impossible in general, and every circle with a repeated speed that attains the three-runner value $L = \mathrm{ML}(1, \frac{N}{2}, N-1)$ lies outside the subtorus. Theorem 10.2 is an explicit instance of the relative spectra of Jain and Kravitz [14].

The obstruction is one phase of a larger family whose values are all three-runner values. For positive integers A, B, C, H with $\gcd(A,B,C) = 1$ put $S = A + C$, $N = S + B$ and

$$U_H(A,B,C) = \{(y_1,y_2,y_3,u) \in \mathbb{T}^4 :\ Ay_1 + By_2 + Cy_3 = 0,\ u = Hy_2\},$$

so that $U(A,B,C) = U_{N/2}(A,B,C)$.

Proposition 10.3 (Coalescence)**.** $\mathrm{ML}(U_H(A,B,C))$ *is the maximum of* $\min(\|y\|, \|z\|, \|Hy\|)$ *over the closed subgroup* $\{(y,z) \in \mathbb{T}^2 : By + Sz = 0\}$. *In particular it depends on* A *and* C *only through* S*, and if* $\gcd(B,S) = 1$ *it equals* $\mathrm{ML}(S,B,HS)$.

Proof. Let $0 \leq d \leq \frac{1}{2}$. If a point of U_H has $G \geq d$, choose representatives $y_i \in [d, 1-d]$ and put $y = y_2$ and $z = (Ay_1 + Cy_3)/S$. Then $z \in [d, 1-d]$, $By + Sz = Ay_1 + By_2 + Cy_3 \in \mathbb{Z}$ and $\|Hy\| = \|u\| \geq d$. Conversely such y, z give the point (z, y, z, Hy) of U_H with $G \geq d$. If $\gcd(B,S) = 1$, the subgroup is the circle $(y,z) = (St, -Bt)$. □

Theorem 10.4 (A phase transition)**.** *Let* $4 \mid N$, $B \geq S$ *and* $H \equiv \frac{N}{2} \pmod N$, *and put* $D^* = \frac{B}{2(HS+B)}$. *Then*

$$\mathrm{ML}(U_H(A,B,C)) = \frac{1}{2} - \min\Big(\frac{1}{N},\ D^*\Big).$$

If $D^* > \frac{1}{N}$, the only maximizers are $\pm p$ with $p = (\frac{1}{2}-\frac{1}{N}, \frac{1}{2}-\frac{1}{N}, \frac{1}{2}-\frac{1}{N}, \frac{1}{2})$, and the argument of Theorem 10.1(ii) applies unchanged. If $D^* < \frac{1}{N}$, the only maximizers are $\pm p^*$ with $p^* = (\frac{1}{2}-D^*, \frac{1}{2}+\frac{S}{B}D^*, \frac{1}{2}-D^*, \frac{1}{2}-D^*)$. If moreover $g = \gcd(B,S)$ divides $\frac{N}{2}$, then the repeated-speed circle $\{(-\frac{B}{g}t, \frac{S}{g}t, -\frac{B}{g}t, \frac{HS}{g}t)\} \subseteq U_H$ contains p^*, and $\mathrm{ML}(U_H(A,B,C)) = \mathrm{ML}(\frac{B}{g}, \frac{S}{g}, \frac{HS}{g})$.

Proof. At a level $\frac{1}{2} - D$ write $y_i = \frac{1}{2} + z_i$ with $|z_i| \le D$. As $\frac{N}{2} \in \mathbb{Z}$, $\lambda \cdot z$ is an integer, and $\frac{H}{2} \in \mathbb{Z}$ gives $u \equiv Hz_2$. If $D < \frac{1}{N}$, then $\lambda \cdot z = 0$, so $B|z_2| \le SD$ and $\|u\| \le \frac{HSD}{B}$, which is less than $\frac{1}{2} - D$ when $D < D^*$. This proves ML $\le \frac{1}{2} - \min(\frac{1}{N}, D^*)$. The point p lies on U_H with $G(p) = \frac{1}{2} - \frac{1}{N}$, because $\lambda \cdot (\frac{1}{2} - \frac{1}{N})(1,1,1) = \frac{N}{2} - 1$ and $H(\frac{1}{2} - \frac{1}{N}) \equiv \frac{1}{2}$. The point p^* lies on U_H with λ-value $\frac{N}{2}$ and $u \equiv \frac{HSD^*}{B} = \frac{1}{2} - D^*$, and $G(p^*) = \frac{1}{2} - D^*$ since $S \le B$. This proves the formula. If $D^* > \frac{1}{N}$, the level $\lambda \cdot z = 0$ is excluded at $D = \frac{1}{N}$, so a maximizer has $\lambda \cdot z = \pm 1$ and is $\mp p$. If $D^* < \frac{1}{N}$, a maximizer has $\lambda \cdot z = 0$ and equality in $B|z_2| \le SD^*$ and in $|Az_1 + Cz_3| \le SD^*$, so it is $\pm p^*$. The displayed circle lies in U_H because $-AB + BS - CB = 0$. On the subgroup of Proposition 10.3 the primitive character $\frac{B}{g}y + \frac{S}{g}z$ takes the value $\frac{N}{2g}$ at the image of p^*, so p^* lies on the identity component, which is the image of this circle, when $g \mid \frac{N}{2}$. The circle has the speeds $\frac{B}{g}, \frac{S}{g}, \frac{HS}{g}$, with $\frac{B}{g}$ repeated. □

For $(A,B,C) = (2,55,3)$, with $N = 60$, $S = 5$ and $g = 5$, the transition is at $H = \frac{B(N-2)}{2S} = 319$. For $H = 30, 90, \ldots, 270$ the value is $\frac{29}{60}$ and no repeated-speed circle in U_H attains it, while for $H = 330$ and 390 the values $\frac{15}{31} = \mathrm{ML}(1, 11, 330)$ and $\frac{195}{401} = \mathrm{ML}(1, 11, 390)$ are attained on $\{y_1 = y_3\}$. So the obstruction of Theorem 10.1 is one phase of the family U_H, and on both sides of the transition the values are three-runner values, $\mathrm{ML}(1, \frac{N}{2}, N-1)$ in the first phase and $\mathrm{ML}(\frac{B}{g}, \frac{S}{g}, \frac{HS}{g})$ in the second when $g \mid \frac{N}{2}$.

11. Near-tight tuples for every number of speeds

In this section $n \ge 3$, $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ and $G(x) = \min_{i \le n} \|x_i\|$, and subtori, Λ_S, covol, coordinate-freeness, T_v and ML are defined as in Section 2. The inequality (3) holds in $\mathbb{T}^n$ with the same proof. We say that LRC(m) holds if every set of m positive integers has loneliness at least $\frac{1}{m+1}$. Since $\mathrm{ML}(V) \ge \mathrm{ML}(V')$ for $V \subseteq V'$, LRC(m) then gives the same bound for every set of at most m positive integers. LRC(m) is known for $m \le 6$ [10, 7, 4, 5, 2], and the computer assisted proofs

[18, 20, 19, 1] cover $m \le 14$. A two-dimensional subtorus $W \subseteq \mathbb{T}^n$ with $\mathrm{ML}(W) = \frac{1}{n}$ is *critical*, and a set of $n-1$ positive integers with loneliness $\frac{1}{n}$ is a *tight instance* for $n-1$ speeds.

Theorem 11.1. *Let $n \ge 3$, assume LRC$(n-1)$ and LRC$(n-2)$, and put $\kappa_n = (n(n-1))^{n-2}$.*

(i) *Every primitive $v \in \mathbb{Z}^n$ with all $v_i \ne 0$ satisfies $\mathrm{ML}(T_v) > \frac{1}{n} - \frac{\kappa_n}{2|v|}$.*

(ii) *Every coordinate-free two-dimensional subtorus $W \subseteq \mathbb{T}^n$ has $\mathrm{ML}(W) \ge \frac{1}{n}$. If $\mathrm{ML}(W) < \frac{1}{n} + \frac{1}{2n\kappa_n}$, then W is critical and $\mathrm{covol}(W) < \kappa_n$. Every critical subtorus contains the circle of a tight instance for $n-1$ speeds.*

(iii) *If $\mathrm{ML}(T_v) < \frac{1}{n}$ and $|v| \ge n\kappa_n^2$, then T_v lies in a critical subtorus of covolume less than κ_n.*

The proof uses three lemmas. The first is a form of [12, Lemma 3.3], which states in the notation $D = \frac{1}{2} - \mathrm{ML}$ that the largest value of D on k-dimensional subtori of $\mathbb{T}^n$ equals the largest value on $(k-\ell)$-dimensional subtori of $\mathbb{T}^{n-\ell}$, and its last statement is [1, Lemma 3.3]. We include a proof because we use the circle it produces.

Lemma 11.2 (Coalescing two coordinates)**.** *Let $S \subseteq \mathbb{T}^n$ be a coordinate-free subtorus of dimension $d \ge 2$, and let ℓ_i be the restriction of the ith coordinate to $E = \mathbb{R}\Lambda_S$. There are $a \ne b$ with ℓ_a, ℓ_b not proportional and $\varepsilon \in \{\pm 1\}$ such that S contains a coordinate-free subtorus of dimension $d-1$ on which $x_a = \varepsilon x_b$. Consequently S contains a coordinate-free circle whose speeds take at most $n-d+1$ distinct absolute values, and $\mathrm{ML}(S) \ge \frac{1}{n-d+2}$ if LRC$(n-d+1)$ holds.*

Proof. The forms ℓ_i are nonzero, take integer values on Λ_S, and span the dual space E^*, which has dimension $d \ge 2$. Let $P \subseteq E^*$ be the plane spanned by two non-proportional forms, and consider the rays $\mathbb{R}_{>0}\sigma\ell_k$ with $\ell_k \in P$ and $\sigma = \pm 1$. They lie on at least two lines, and every line meets both open half-planes bounded by any other line. Hence two consecutive rays in the cyclic order around the origin, say $\mathbb{R}_{>0}\sigma\ell_a$ and $\mathbb{R}_{>0}\tau\ell_b$, lie on different lines and make an angle less than π. The form $\phi = \sigma\ell_a + \tau\ell_b$ lies in the open cone between them, which contains no ray, so ϕ is proportional to no ℓ_k (such an ℓ_k would lie in P). Put $\varepsilon = -\sigma\tau$, so that $\phi = \sigma(\ell_a - \varepsilon\ell_b)$, and let $H = \ker\phi$. It is a rational hyperplane of E, so the subtorus S_H with $\mathbb{R}\Lambda_{S_H} = H$ has dimension $d-1$, it lies in S, and $x_a = \varepsilon x_b$ on S_H. A coordinate vanishes on S_H exactly when ℓ_k vanishes on H, that is, when ℓ_k is proportional to ϕ, so S_H is coordinate-free.

Call two coordinates equivalent on a subtorus if they agree there up to sign. On S_H the classes of a and b merge, so the number of

classes drops by at least one. Starting from at most n classes on S and applying the first part $d-1$ times gives a coordinate-free circle $T \subseteq S$ with at most $n-d+1$ classes, that is, whose speeds take at most $n-d+1$ distinct absolute values. Then $\mathrm{ML}(S) \geq \mathrm{ML}(T) \geq \frac{1}{n-d+2}$ by $\mathrm{LRC}(n-d+1)$. □

Lemma 11.3 (Density in $\mathbb{T}^n$)**.** *Let S be a k-dimensional subtorus of $\mathbb{T}^n$ with $1 \leq k < n$, let $E = \mathbb{R}\Lambda_S$, let $\pi : \mathbb{R}^n \to E^\perp$ be the orthogonal projection, and let $\lambda_1 \leq \cdots \leq \lambda_{n-k}$ be the successive minima of the lattice $\pi(\mathbb{Z}^n)$ with respect to the maximum norm.*

(a) $\lambda_1 \cdots \lambda_{n-k} \leq 1/\operatorname{covol}(S)$.
(b) *Let $w_1, \ldots, w_j \in \mathbb{Z}^n$ have linearly independent projections $\pi(w_i)$. The subtorus S' with $\mathbb{R}\Lambda_{S'} = E + \mathbb{R}w_1 + \cdots + \mathbb{R}w_j$ has dimension $k+j$, and every point of S' lies within maximum norm distance $\frac{1}{2}\sum_i |\pi(w_i)|_\infty$ of a point of S. In particular* $\mathrm{ML}(S) \geq \mathrm{ML}(S') - \frac{1}{2}\sum_i |\pi(w_i)|_\infty$.

Proof. As in Lemma 3.1(a), $\pi(\mathbb{Z}^n)$ is a lattice of rank $n-k$ in $E^\perp$ with covolume $1/\operatorname{covol}(S)$. The body $K = \{z \in E^\perp : |z|_\infty \leq 1\}$ has volume at least 2^{n-k} by Vaaler's theorem [21], and Minkowski's second theorem [6, Chapter VIII] gives $\lambda_1 \cdots \lambda_{n-k} \operatorname{vol}(K) \leq 2^{n-k} \operatorname{covol}(\pi(\mathbb{Z}^n))$, which is (a). For (b), $E + \mathbb{R}w_1 + \cdots + \mathbb{R}w_j = E + \mathbb{R}\pi(w_1) + \cdots + \mathbb{R}\pi(w_j)$ has dimension $k+j$. Write $w_i = \pi(w_i) + e_i$ with $e_i \in E$, represent a point of S' by $y = s + \sum_i b_i \pi(w_i)$ with $s \in E$, and let c_i be an integer nearest to b_i. Then $y - \sum_i c_i w_i = (s - \sum_i c_i e_i) + \sum_i (b_i - c_i)\pi(w_i)$ represents the same point, its first summand represents a point of S, and its second summand has maximum norm at most $\frac{1}{2}\sum_i |\pi(w_i)|_\infty$. The last claim follows from (3). □

Lemma 11.4 (Corner lemma for planes)**.** *Let $W \subseteq \mathbb{T}^n$ be a coordinate-free two-dimensional subtorus. Then $\mathrm{ML}(W) = p/q$ with integers $p \geq 1$ and $1 \leq q \leq 2\operatorname{covol}(W)$.*

Proof. Let b, b' be a basis of Λ_W and $r_i = (b_i, b_i') \in \mathbb{Z}^2$. Then W is the image of $\mathbb{R}^2$ under $y \mapsto (r_i \cdot y)_i$, the r_i span $\mathbb{R}^2$, and the minors $D_{ij} = \det(r_i, r_j)$ satisfy $\sum_{i<j} D_{ij}^2 = \operatorname{covol}(W)^2$ by the Cauchy–Binet formula, so $\operatorname{covol}(W) \geq 1$. Let $f(y) = \min_i \|r_i \cdot y\|$, let y^* be a maximizer and $\mu = f(y^*)$, which is positive because W contains a coordinate-free circle. If $\mu = \frac{1}{2}$ we are done. Otherwise let I be the set of i with $\|r_i \cdot y^*\| = \mu$. For $i \in I$ there are $\sigma_i \in \{\pm 1\}$ and $k_i \in \mathbb{Z}$ with $\|r_i \cdot y\| = \sigma_i(r_i \cdot y - k_i)$ near y^*, and $f(y) = \min_{i \in I} \sigma_i(r_i \cdot y - k_i)$ near y^*. Hence (y^*, μ) is a local, and therefore global, maximum point of the linear function ν on the convex polyhedron $P = \{(y, \nu) : \sigma_i(r_i \cdot y - k_i) \geq \nu,\ i \in I\}$. A line

contained in P has a direction (δ, η) with $\sigma_i r_i \cdot \delta = \eta$ for $i \in I$, and $\eta = 0$ because ν is bounded on P.

If the r_i with $i \in I$ span $\mathbb{R}^2$, then P contains no line, so ν attains μ at a vertex of P, where three constraints i, j, m with linearly independent rows $(\sigma_i r_i, -1)$ are active. By Cramer's rule $\mu \in \frac{1}{\Delta}\mathbb{Z}$, where Δ is the absolute value of the determinant of these rows, and expanding along the last column gives $\Delta \le |D_{ij}| + |D_{im}| + |D_{jm}| \le \sqrt{3}\,\mathrm{covol}(W)$. Otherwise $r_i = c_i w$ for $i \in I$, with a primitive $w \in \mathbb{Z}^2$ and integers $c_i \neq 0$. Then the constraints depend on y only through $z = w \cdot y$, and the maximum of ν on the polygon $\{(z, \nu) : \sigma_i(c_i z - k_i) \ge \nu,\ i \in I\}$, which contains no line, is attained at a vertex, where two constraints i, j with $\sigma_i c_i \neq \sigma_j c_j$ are active. So $\mu \in \frac{1}{|\sigma_i c_i - \sigma_j c_j|}\mathbb{Z}$. Some r_m is not parallel to w, and $|c_i| \le |c_i \det(w, r_m)| = |D_{im}|$ and likewise $|c_j| \le |D_{jm}|$, so $|\sigma_i c_i - \sigma_j c_j| \le \sqrt{2}\,\mathrm{covol}(W)$. □

Proof of Theorem 11.1. By Lemma 11.2, coordinate-free subtori of dimension two have loneliness at least $\frac{1}{n}$, and those of dimension three have loneliness at least $\frac{1}{n-1}$.

(ii) Let W be coordinate-free of dimension two with $\mathrm{ML}(W) < \frac{1}{n} + \frac{1}{2n\kappa_n}$, and let $A = \mathrm{covol}(W)$. By Lemma 11.3(a) with $k = 2$ there is $w \in \mathbb{Z}^n$ with $0 < |\pi(w)|_\infty = \lambda_1 \le A^{-1/(n-2)}$, and Lemma 11.3(b) gives a coordinate-free three-dimensional subtorus $S' \supseteq W$ with $\frac{1}{n-1} \le \mathrm{ML}(S') \le \mathrm{ML}(W) + \frac{1}{2}\lambda_1$. Hence

$$A^{-1/(n-2)} \ \ge\ \lambda_1 \ \ge\ \frac{2}{n-1} - 2\,\mathrm{ML}(W) \ >\ \frac{2}{n(n-1)} - \frac{1}{n\kappa_n} \ \ge\ \frac{1}{n(n-1)},$$

since $\kappa_n \ge n - 1$, so $A < \kappa_n$. By Lemma 11.4, $\mathrm{ML}(W) = p/q$ with $q \le 2A < 2\kappa_n$. If $\mathrm{ML}(W) \neq \frac{1}{n}$, then $\mathrm{ML}(W) - \frac{1}{n} = \frac{np-q}{nq} \ge \frac{1}{nq} > \frac{1}{2n\kappa_n}$, a contradiction. If W is critical, Lemma 11.2 gives a coordinate-free circle $T \subseteq W$ whose speeds take at most $n-1$ distinct absolute values, and $\mathrm{ML}(T) \le \mathrm{ML}(W) = \frac{1}{n}$. By $\mathrm{LRC}(n-2)$ there are exactly $n-1$ values, and by $\mathrm{LRC}(n-1)$ we get $\mathrm{ML}(T) = \frac{1}{n}$.

(i) Suppose that $\mathrm{ML}(T_v) < \frac{1}{n}$. Apply Lemma 11.3 to $S = T_v$, for which $k = 1$ and $\mathrm{covol}(S) = |v|$, and choose $w_1, w_2 \in \mathbb{Z}^n$ with linearly independent projections and $|\pi(w_i)|_\infty = \lambda_i$. Part (b) gives coordinate-free subtori $W \supseteq T_v$ of dimension two and $S_3 \supseteq T_v$ of dimension three with

$$\mathrm{ML}(T_v) \ge \mathrm{ML}(W) - \tfrac{1}{2}\lambda_1 \ge \tfrac{1}{n} - \tfrac{1}{2}\lambda_1,$$
$$\mathrm{ML}(T_v) \ge \mathrm{ML}(S_3) - \tfrac{1}{2}(\lambda_1 + \lambda_2) \ge \tfrac{1}{n-1} - \lambda_2.$$

The second inequality gives $\lambda_i \geq \lambda_2 > \frac{1}{n-1} - \frac{1}{n} = \frac{1}{n(n-1)}$ for $i \geq 2$, so part (a) gives $\lambda_1 < \kappa_n/|v|$, and the first inequality gives (i).

(iii) For W as in the proof of (i), $\mathrm{ML}(W) \leq \mathrm{ML}(T_v) + \frac{1}{2}\lambda_1 < \frac{1}{n} + \frac{\kappa_n}{2|v|} \leq \frac{1}{n} + \frac{1}{2n\kappa_n}$, and (ii) applies. □

The hypotheses of Theorem 11.1 hold for $3 \leq n \leq 7$, and for $n \leq 15$ by the computer assisted proofs cited above. They are the hypotheses of Cordella's reduction lemma [8, Lemma 2.4], and part (iii) is its effective form: a near-tight n-tuple on no critical subtorus has $|v| < n\kappa_n^2$. This answers [8, Question 6.7] for every n. The step of [8] that is not effective is a gap above $\frac{1}{n}$ in the values of two-dimensional subtori, which comes from the fact that, in the notation $D = \frac{1}{2} - \mathrm{ML}$, these values have only upper accumulation points [12, Theorem 1.4]. Part (ii) makes the gap explicit and bounds the covolume of the critical subtori. Its last statement explains why tight instances for $n-1$ speeds govern the near-tight n-tuples. For $n = 4$ the circle of $\{1, 2, 3\}$ lies on U^1 (at $a = b$) and on U^2 (at $x = y$), and for $n = 5$ Cordella combines his reduction with Chen's two tight quadruples $\{1, 2, 3, 4\}$ and $\{1, 3, 4, 7\}$. In contrast with Section 10, at the lowest value $\frac{1}{n}$ a circle with a repeated speed always attains $\mathrm{ML}(W)$.

Part (i) says that the values of ML below $\frac{1}{n}$ approach $\frac{1}{n}$ at rate $1/|v|$. For $\eta > 0$ the values at most $\frac{1}{n} - \eta$ come from tuples with $|v| < \kappa_n/(2\eta)$, so they form an explicit finite set, and the exponent of $|v|$ is best possible because $\mathrm{ML}(1, 2, 3, 4k) = \frac{1}{4} - \frac{1}{16k+4}$ by Theorem 1.1. Giri and Kravitz [12, Section 7] showed effectively that a circle of length $|v|$ is ε-dense in a two-dimensional subtorus once $|v| \geq c_n \varepsilon^{1-n}$ with an explicit c_n, which gives a deficit of order $|v|^{-1/(n-1)}$. The exponent 1 comes from the second minimum λ_2, which $\mathrm{LRC}(n-2)$ controls. For the conjecture itself, (i) reduces the case of n speeds to $|v| < n(n+1)\kappa_n/2$. This does not improve the bound $\binom{n+1}{2}^{n-1}$ of Malikiosis, Santos and Schymura [16], and Allikvere [1, Theorem 3.8] bounds counterexamples much more sharply by applying $\mathrm{LRC}(m)$ for all $m < n$ along a whole flag of subtori containing T_v. The content of (i) is the range between $\frac{1}{n+1}$ and $\frac{1}{n}$, where counterexample bounds say nothing. The constants are far from what an exhaustive search can handle: $4\kappa_4^2 = 82{,}944$ against the bound 512 of Proposition 6.2, and $5\kappa_5^2 = 3.2 \cdot 10^8$, $6\kappa_6^2 \approx 3.9 \cdot 10^{12}$. So for five and six speeds the theorems of [8] now hold for all near-tight tuples outside an explicit finite set, and checking that set needs sharper constants, as Propositions 5.2 and 6.2 provide for $n = 4$.

11.1. **Five speeds.** For five speeds the general constants can be replaced by an exact statement up to $\frac{2}{9}$, because Theorem 1.1

identifies the circles with a repeated speed that a subtorus of small loneliness must contain. For a quadruple u write $T_u \times \mathbb{T} = \{(u_1 t, u_2 t, u_3 t, u_4 t, s)\} \subseteq \mathbb{T}^5$. Its loneliness is $\mathrm{ML}(u)$, since the last coordinate can be put at $\frac{1}{2}$.

Theorem 11.5. *Let $W \subseteq \mathbb{T}^5$ be a coordinate-free two-dimensional subtorus with $\mathrm{ML}(W) < \frac{2}{9}$. Then, up to signed coordinate permutations, $W = T_{(1,2,3,4)} \times \mathbb{T}$ or $W = T_{(1,3,4,7)} \times \mathbb{T}$, and $\mathrm{ML}(W) = \frac{1}{5}$.*

Proof. By Lemma 11.2, W contains a coordinate-free circle T whose speeds take at most four distinct absolute values, and $\mathrm{ML}(T) \le \mathrm{ML}(W) < \frac{2}{9}$. At most three values would give $\mathrm{ML}(T) \ge \frac{1}{4}$ by the conjecture for three speeds, so there are exactly four, and by Corollary 1.3(iii) they are a multiple of $\{1,2,3,4\}$ or of $\{1,3,4,7\}$. The same holds for every coordinate-free circle in W with a repeated absolute speed. After a signed coordinate permutation, T is the circle of one of the eight vectors

$$(1,1,2,3,4), \quad (1,2,2,3,4), \quad (1,2,3,3,4), \quad (1,2,3,4,4),$$
$$(1,1,3,4,7), \quad (1,3,3,4,7), \quad (1,3,4,4,7), \quad (1,3,4,7,7),$$

call it q. Since q is primitive and lies in the primitive lattice Λ_W, we have $\Lambda_W = \mathbb{Z}q + \mathbb{Z}c$ for some $c \in \mathbb{Z}^5$. Replacing c by $\pm c + mq$ we may assume $c_1 = 0$, as $q_1 = 1$, and that the first nonzero entry of c is positive. The minors $c_k = q_1 c_k - q_k c_1$ show that $\mathbb{Z}q + \mathbb{Z}c$ is primitive exactly when $\gcd(c) = 1$.

Next we bound $\mathrm{covol}(W)$. Apply Lemma 11.3 to $S = W$, with $k = 2$ and successive minima $\lambda_1 \le \lambda_2 \le \lambda_3$. Part (b) gives coordinate-free subtori containing W of dimensions three, four and five, with loneliness at least $\frac{1}{4}$, $\frac{1}{3}$ and $\frac{1}{2}$ by Lemma 11.2 and the conjecture for three and two speeds. Hence $\lambda_1 > 2(\frac{1}{4} - \frac{2}{9}) = \frac{1}{18}$, $\lambda_1 + \lambda_2 > 2(\frac{1}{3} - \frac{2}{9}) = \frac{2}{9}$ and $\lambda_1 + \lambda_2 + \lambda_3 > 2(\frac{1}{2} - \frac{2}{9}) = \frac{5}{9}$. So $\lambda_2 > \frac{1}{9}$ and $\lambda_3 > \frac{5}{27}$, and part (a) gives $\mathrm{covol}(W) < \frac{4374}{5}$, that is

$$\mathrm{covol}(W)^2 = \sum_{i<j} (q_i c_j - q_j c_i)^2 \le 765{,}275.$$

A circle of W with a repeated absolute speed is the image of a line in the direction orthogonal to $r_i - \varepsilon r_j$ for some $i < j$ and $\varepsilon = \pm 1$, where $r_k = (q_k, c_k)$ are the coordinate rows of the parametrization $(t, s) \mapsto tq + sc$. For each of the eight vectors q, the program `g5enum2` (Section 12) enumerates every normalized c in this range and tests all of these directions. It finds that $W = \mathbb{Z}q + \mathbb{Z}c$ satisfies the condition of the first paragraph only for 16 pairs (q, c), two for each q, and in each of them one of the two coordinates where q has a repeated entry is a

free coordinate of W while the other four coordinates are the circle of $\{1,2,3,4\}$ or $\{1,3,4,7\}$. So W is one of the two products, and both have loneliness $\frac{1}{5}$. □

Corollary 11.6. (i) *The accumulation points of the set of values* $\mathrm{ML}(V)$*, over all sets* V *of five positive integers, that lie in* $(0,\frac{2}{9})$ *consist of* $\frac{1}{5}$ *alone. In the notation* $D=\frac{1}{2}-\mathrm{ML}$ *this is the case* $n=5$ *of* [12, Conjecture 1.5] *on* $(\frac{5}{18},\frac{1}{2})$.

(ii) *Every primitive five-speed tuple* v *with* $\mathrm{ML}(v)<\frac{1}{5}$ *and* $|v|\geq 33{,}750$ *is, up to order,* $\{a,2a,3a,4a,b\}$ *or* $\{a,3a,4a,7a,b\}$ *with* $\gcd(a,b)=1$*. In particular the critical subtori for five speeds are the two products of Theorem 11.5.*

Proof. (i) By [12, Theorem 1.4], $\mathcal{S}(4)\subseteq \mathrm{acc}(\mathcal{S}(5))\subseteq \mathcal{S}_2(5)$. The two conventions are reconciled as in the proof of Corollary 9.7: a tuple with a repeated speed has the value of a set of at most four speeds, and by Corollary 1.3 the only such value below $\frac{2}{9}$ is $\frac{1}{5}$. Theorem 11.5 gives $\mathcal{S}_2(5)\cap(0,\frac{2}{9})=\{\frac{1}{5}\}$, and $\frac{1}{5}=\mathrm{ML}(1,2,3,4)$.

(ii) Apply Lemma 11.3 to $S=T_v$ as in the proof of Theorem 11.1(i). With S_{j+1} the subtorus of dimension $j+1$ built from the first j minima, $\mathrm{ML}(S_{j+1})\geq \frac{1}{4},\frac{1}{3},\frac{1}{2}$ for $j=2,3,4$, so $\mathrm{ML}(v)<\frac{1}{5}$ gives $\lambda_1+\lambda_2>\frac{1}{10}$, $\lambda_1+\lambda_2+\lambda_3>\frac{4}{15}$ and $\lambda_1+\cdots+\lambda_4>\frac{3}{5}$. Hence $\lambda_2\lambda_3\lambda_4>\frac{1}{20}\cdot\frac{4}{45}\cdot\frac{3}{20}=\frac{1}{1500}$ and $\lambda_1<1500/|v|$. The two-dimensional subtorus $W=S_2\supseteq T_v$ has $\mathrm{ML}(W)\leq \mathrm{ML}(v)+\frac{1}{2}\lambda_1<\frac{1}{5}+\frac{750}{|v|}\leq\frac{2}{9}$, so W is one of the products of Theorem 11.5, and $v\in\Lambda_W$ gives the two forms. The last statement follows because a critical subtorus has loneliness $\frac{1}{5}<\frac{2}{9}$. □

Part (ii) is the effective form of Cordella's reduction for five speeds [8], with the critical subtori determined and the bound $n\kappa_n^2=3.2\cdot10^8$ of Theorem 11.1(iii) replaced by 33,750. For four speeds the corresponding statement is Proposition 6.2, where the second critical subtorus U^2 is not a product. For five speeds no such subtorus exists below $\frac{2}{9}$.

12. Computations and reproducibility

All programs are written in C with exact 64-bit integer arithmetic (the Hermite normal form enumeration uses floating point square roots only to set loop bounds, with a margin of one on each side, and every candidate is then tested exactly), and all claims that depend on them are re-verified by a Python script using only the standard library and exact fractions. The programs are the following.

Exact loneliness of a two-dimensional subtorus. For forms $\ell_i(X,Y)=s_iX+r_iY$, the maximum of $\min_i\|\ell_i\|$ over $\mathbb{T}^2$ is attained at a point where three signed linear branches $\sigma_i(\ell_i-k_i)$ are equal, or where two

proportional forms are at a common level and a third form is at a half-integer, or where two independent forms are at half-integers. Every such point is the solution of a 3×3 linear system with integer right-hand side in a bounded range. The program enumerates all of them, evaluates the minimum exactly at each, and returns the best point. It agrees with an independent implementation and with a fine grid search on several hundred random subtori with small coefficients. In the proof only the returned points are used, as certificates.

Enumeration of two-dimensional subtori. Hermite normal forms for case (i) of Lemma 5.1, and primitive $n \in \mathbb{Z}^3$ for case (ii). Running times are ten and three seconds. The enumeration of Proposition 9.5, with covolume up to 57, the special-subtorus test and the threshold $1/3$, runs in 54 seconds. The enumeration of Proposition 9.14, with squared covolume up to 11,331, the tests for the 40 special characters and the 176 characters with $|m|_1 = 5$ up to sign, and the threshold $\frac{2}{5}$, ran in two parallel jobs of 21 minutes. When the grids $Q \in \{2, 10, 20, 30\}$ give no point with $G \geq \frac{2}{5}$, the program calls the exact maximization, in a second implementation of the same candidate enumeration in which the integer offsets of a base pair of independent forms run only over a fundamental domain of the lattice they generate. This happened for 44,839,680 lattices, of which 18,960 had $\mathrm{ML} < \frac{2}{5}$. The two implementations agree on all 916,492 coordinate-free subtori of covolume at most 20, on 19,300 further subtori with larger coefficients, and on all 18,960 survivors. The formula in the proof of Lemma 9.13 was compared with the C maximizer for all subtori U_n of the five tori with $|n|_\infty \leq 20$, 142,547 subtori, with no difference, and every value in $[\frac{1}{3}, \frac{2}{5})$ among them lies in $\mathcal{R}$.

Section 10. The exact maximizer confirms Theorem 10.1(i) on 255 random admissible subtori with $N \leq 126$. The loneliness of every repeated-speed circle was computed by Lemma 2.1 for 10,958 admissible subtori with $N \leq 196$, 58 of them with all three greatest common divisors above 1, and for the examples $m \leq 4$, $K \leq 2$, in agreement with parts (ii) and (iii). The formula of Theorem 10.2 was checked for all $4 \mid N \leq 124$ and $\ell \leq 6$, and the comparison for $(N, \ell) = (60, 61)$ and $(120, 121)$. Proposition 10.3 was checked on 300 random subtori with $\gcd(B, S) = 1$ against $\mathrm{ML}(S, B, HS)$, and its independence of A and C on 1,440 groups of parameters. Theorem 10.4 was checked on 5,460 subtori with $N \leq 100$ and $H = \frac{N}{2} + jN$, $j \in \{0, 1, 2, 5, 9\}$, including the triple values of the second phase in all 3,651 cases there with $g \mid \frac{N}{2}$.

Section 11.1. The program `g5enum2` runs through the vectors $c = (0, c_2, \ldots, c_5)$ with $\sum_{i<j}(q_i c_j - q_j c_i)^2 \leq 765{,}275$ by nested loops whose

ranges come from the partial sums of this positive definite quadratic form, keeps those with $\gcd(c) = 1$ and first nonzero entry positive, and tests all directions $r_i \mp r_j$. For the eight vectors q together it visits 65,989,492,206 vectors, of which 30,485,136,454 are normalized, in two parallel jobs of 47 and 40 minutes, and exactly 16 pass. They are the products of Theorem 11.5, with squared covolume 30 or 75. A Python search over a box, without the pruning, reproduces the number of normalized vectors and the survivors for all eight q with the bound 3,000 in place of 765,275. The program `torusmlN` is the exact maximizer `torusml2` extended to any number of forms. It agrees with `torusml2` on 289 random inputs with four forms and with a separate Python implementation on 80 inputs with five forms, and it confirms $\mathrm{ML} = \frac{1}{5}$ for the 16 survivors. It also computes $\mathrm{ML}(\mathbb{Z}q+\mathbb{Z}c)$ for all 8,960 normalized c with $|c_i| \le 3$: the 14 with $\mathrm{ML} < \frac{2}{9}$ all pass the test and have $\mathrm{ML} = \frac{1}{5}$. On 20,000 random primitive planes in $\mathbb{Z}^5$ with coordinates in $[-3, 3]$ it finds no value in $(\frac{1}{5}, \frac{2}{9})$.

Exhaustive search. All primitive quadruples with $|v| < 512$, in two parallel runs of under one minute each.

Verification script. It checks the formulas of Theorems 8.2 and 8.4 against the exact corner computation for all coprime $x, y \le 40$ and $b \le 10$, $a \le 60$, the 21 classes of characters and the certificates of Table 1, the 123 certificates of Section 5 including the covolumes and the coordinate-freeness, the classification of all primitive quadruples with maximum speed at most 60 by an independent pure Python enumeration, the absence of $m = 3$ and the witnesses for all other m, the formulas of Lemmas 9.2 and 9.4 against an exact two-dimensional maximizer written in Python (524 subtori of W^1 with $|n_1| \le 6$, $|n_2|, |n_3| \le 3$, 458 subtori of W^2 with $|n_1|, |n_2| \le 4$, $|n_3| \le 3$, covering all three cases of Lemma 9.4, and $F(n_1, n_2)$ for $|n_i| \le 4$), the by-product on triples for maximum speed at most 80, and the output file of the exhaustive search, including 200 random exact re-computations. Larger ranges were checked with the C maximizer: 3,356 subtori of W^1 and 3,354 of W^2 with $|n_1| \le 12$, $|n_2|, |n_3| \le 6$, the 624 subtori $F(n_1, n_2)$ with $|n_i| \le 12$, and the by-product on all 1,098,601 primitive triples with maximum speed at most 200, of which 4,639 have $\mathrm{ML} < 1/3$. For Section 9.1 the script checks the arithmetic of Lemma 9.11, the five types of characters with $|m|_1 = 5$ and their parametrizations, Lemma 9.12 at all points of the grid $(\frac{1}{60}\mathbb{Z})^3$ for three values of ε, the formula in the proof of Lemma 9.13 against its own exact maximizer for all U_n with $|n|_\infty \le 3$ in the five tori, the formulas for $\mathrm{ML}(2, 3, 5j)$ and $\mathrm{ML}(1, 4, 5j + 3)$ for $j \le 40$, the two descriptions of $\mathcal{R}$, the example with value $\frac{43}{103}$ at the

level $\frac{3}{7}$, and the example of Remark 9.16. For Section 10 it checks $\mathrm{ML}(1, \frac{N}{2}, N-1)$ for $4 \mid N \le 200$, five subtori $U(A,B,C)$ with $N \le 16$ and all their repeated-speed circles, Theorem 10.2 for $N \le 60$, $\ell \le 3$, the comparison in $U(2,55,3)$ with $\ell = 61$, Proposition 10.3 on small cases, Theorem 10.4 on both sides of the transition for $N = 12$, and the two values $\frac{15}{31}$ and $\frac{195}{401}$. For the 18,960 survivors of Proposition 9.14 it checks the witness points, the squared covolumes and the values, reduces them to the 141 orbits under signed permutations, and recomputes the value of each orbit with its own exact maximizer.

Appendix A. Tables

m	point $x \in W_m$	$G(x)$
$(0,0,1,-1)$	$(2/5,2/5,2/5,2/5)$	$2/5$
$(1,-1,-1,-1)$	$(2/5,2/5,2/5,3/5)$	$2/5$
$(0,1,-2,-1)$	$(2/5,2/5,2/5,3/5)$	$2/5$
$(1,-2,-1,-1)$	$(2/5,3/5,3/5,3/5)$	$2/5$
$(0,1,-2,-2)$	$(2/5,2/5,3/5,3/5)$	$2/5$
$(0,0,1,-3)$	$(2/5,2/5,1/2,1/2)$	$2/5$
$(1,-2,-2,-1)$	$(2/5,2/5,3/5,2/5)$	$2/5$
$(0,1,-3,-1)$	$(2/5,2/5,3/5,3/5)$	$2/5$
$(1,-3,-1,-1)$	$(2/5,2/5,3/5,3/5)$	$2/5$
$(0,0,2,-3)$	$(2/5,2/5,2/5,3/5)$	$2/5$
$(1,-2,-2,-2)$	$(2/5,2/5,2/5,2/5)$	$2/5$
$(0,1,-3,-2)$	$(2/5,2/5,2/5,3/5)$	$2/5$
$(1,-3,-2,-1)$	$(2/5,2/5,2/5,2/5)$	$2/5$
$(0,0,1,-4)$	$(2/5,2/5,2/5,3/5)$	$2/5$
$(0,2,-3,-2)$	$(2/5,2/5,3/5,1/2)$	$2/5$
$(0,1,-4,-1)$	$(2/5,2/5,1/2,2/5)$	$2/5$
$(1,-3,-2,-2)$	$(2/5,3/5,2/5,2/5)$	$2/5$
$(0,1,-3,-3)$	$(2/5,2/5,2/5,2/5)$	$2/5$
$(1,-4,-1,-1)$	$(2/5,2/5,2/5,2/5)$	$2/5$
$(0,0,1,-2)$	W^1 up to symmetry	$\mathrm{ML} = 1/3$
$(0,1,-1,-1)$	W^2 up to symmetry	$\mathrm{ML} = 1/3$

Table 1. The 21 classes of primitive $m \in \mathbb{Z}^4$ with at least two nonzero entries and $|m|^2 \le 19$, up to signed permutations, with certificates for Lemma 4.2. In each row $m \cdot x \in \mathbb{Z}$.

ML(U)	covol^2	parametrization $(X, Y) \mapsto$	maximizer
1/4	9	$(X, Y, -X-2Y, -X-Y)$	$(3/4, 3/4)$
1/4	14	$(X, Y, -3Y, -2Y)$	$(3/4, 3/4)$
2/7	21	$(X, Y, -3X-3Y, -X-Y)$	$(5/7, 5/7)$
2/7	26	$(X, Y, -4Y, -3Y)$	$(5/7, 4/7)$
2/7	27	$(X, Y, -2X-4Y, -X-Y)$	$(5/7, 5/7)$
2/7	38	$(X, 2Y, -5Y, -3Y)$	$(5/7, 6/7)$
2/7	41	$(X, Y, -6Y, -2Y)$	$(5/7, 5/7)$
2/7	62	$(X, Y, -6Y, -5Y)$	$(5/7, 5/7)$
3/10	45	$(X, Y, -4X-5Y, -X-Y)$	$(7/10, 7/10)$
3/10	57	$(X, Y, -3X-6Y, -X-Y)$	$(7/10, 7/10)$
3/10	74	$(X, 3Y, -7Y, -4Y)$	$(7/10, 9/10)$
3/10	86	$(X, Y, -9Y, -2Y)$	$(7/10, 7/10)$
3/10	146	$(X, Y, -9Y, -8Y)$	$(7/10, 7/10)$
4/13	75	$(X, Y, -6X-6Y, -X-Y)$	$(9/13, 9/13)$
4/13	81	$(X, Y, -5X-7Y, -X-Y)$	$(9/13, 9/13)$
4/13	86	$(X, Y, -7Y, -6Y)$	$(9/13, 8/13)$
4/13	98	$(X, 3Y, -8Y, -5Y)$	$(9/13, 7/13)$
4/13	99	$(X, Y, -4X-8Y, -X-Y)$	$(9/13, 9/13)$
4/13	122	$(X, 4Y, -9Y, -5Y)$	$(9/13, 12/13)$
4/13	149	$(X, Y, -12Y, -2Y)$	$(9/13, 9/13)$
4/13	158	$(X, 3Y, -10Y, -7Y)$	$(9/13, 3/13)$
4/13	206	$(X, 2Y, -11Y, -9Y)$	$(9/13, 11/13)$
4/13	266	$(X, Y, -12Y, -11Y)$	$(9/13, 9/13)$
5/16	117	$(X, Y, -7X-8Y, -X-Y)$	$(11/16, 11/16)$
5/16	129	$(X, Y, -6X-9Y, -X-Y)$	$(11/16, 11/16)$
5/16	134	$(X, 2Y, -9Y, -7Y)$	$(11/16, 13/16)$
5/16	153	$(X, Y, -5X-10Y, -X-Y)$	$(11/16, 11/16)$
5/16	182	$(X, 5Y, -11Y, -6Y)$	$(11/16, 15/16)$
5/16	230	$(X, Y, -15Y, -2Y)$	$(11/16, 11/16)$
5/16	278	$(X, 3Y, -13Y, -10Y)$	$(11/16, 9/16)$
5/16	422	$(X, Y, -15Y, -14Y)$	$(11/16, 11/16)$

TABLE 2. The 31 coordinate-free two-dimensional subtori of $\mathbb{T}^4$ with ML $\leq 5/16$, up to signed coordinate permutations, with a maximizer in the given coordinates. The first two are U^2 and U^1 up to symmetry. The subtori of the form $(X, cY, \ldots)$ with the last three coordinates depending on Y only are products of a three-runner circle with a full coordinate circle, and their values are the three-runner values $s/(3s+1)$.

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